

IEOR E4004

Optimization Models and Methods

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Outline

- Solving Linear Programs
 - Graphical Solution Method
- Introduction to Solvers
 - MS Excel Solver Example
 - Gurobi Example

Solving Linear Programs

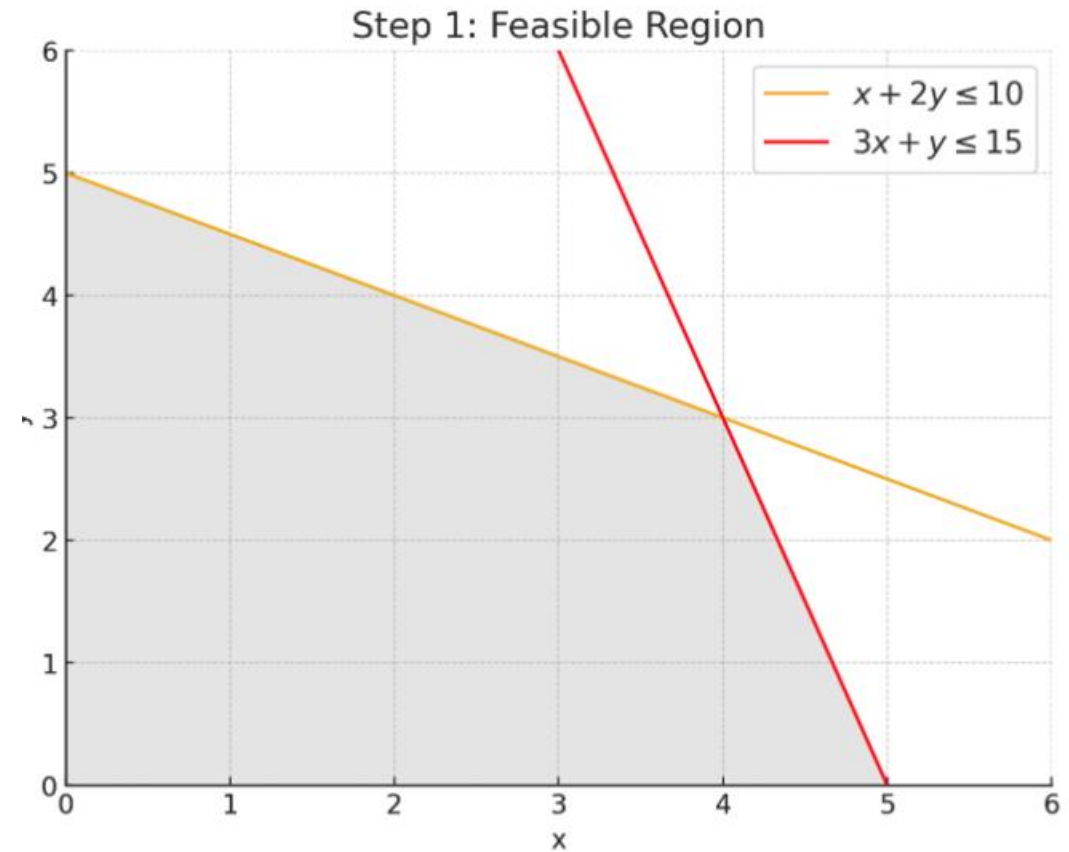
Graphical Solution Method

Solving a Linear Program Graphically

Step 1: Graph the region that satisfies all of the constraints.

Example:

- Constraints:
 $x + 2y \leq 10$
 $3x + y \leq 15$
 $x \geq 0$
 $y \geq 0$

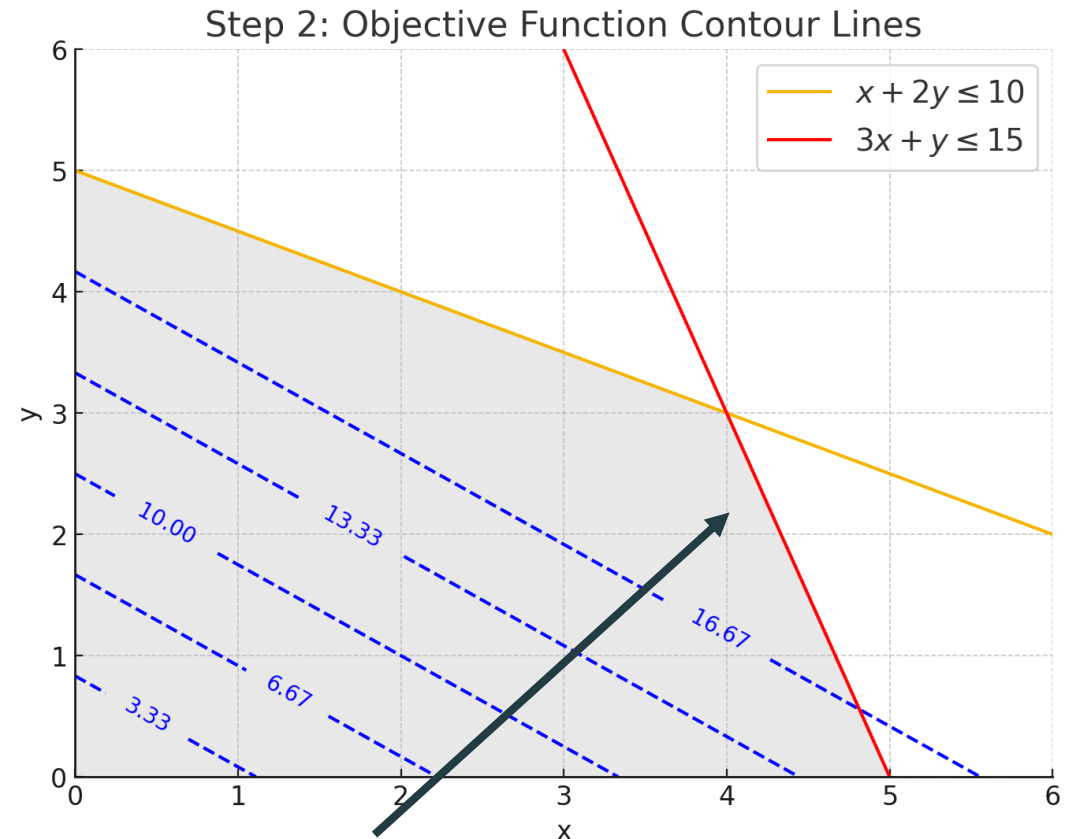


Solving a Linear Program Graphically

Step 2: Draw a contour line of the objective function and determine which direction improves the objective function. A second contour line can be used to determine which direction is improving.

Objective function: $z = 3x + 4y$

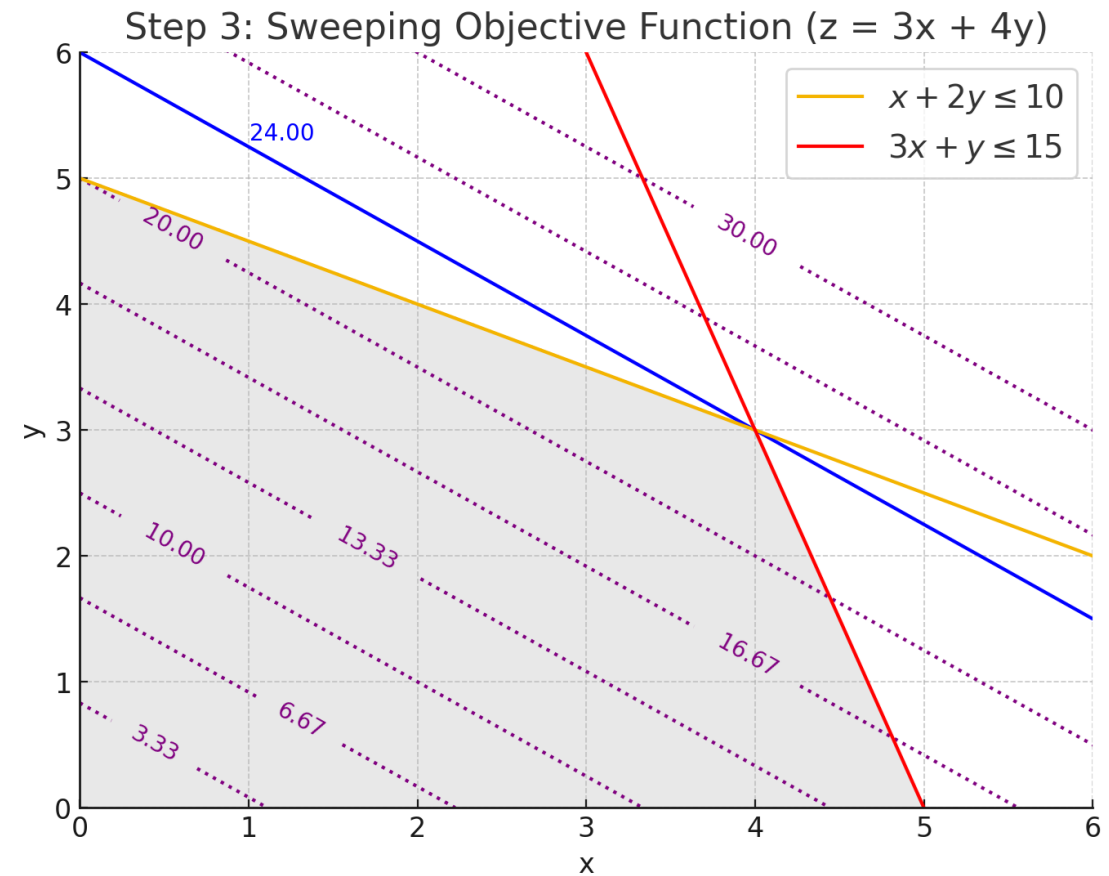
- The improving direction is towards higher values as we aim to maximize z



Solving a Linear Program Graphically

Step 3: Sweep the contour line in the improving direction, generating parallel lines, until it is about to leave the region. The last line intersecting the region has the optimal objective function value. The point(s) where this line intersects the region is optimal.

- Here, solid blue line represents $z=24$, which is the maximum value achievable within the feasible region.



Solving a Linear Program Graphically

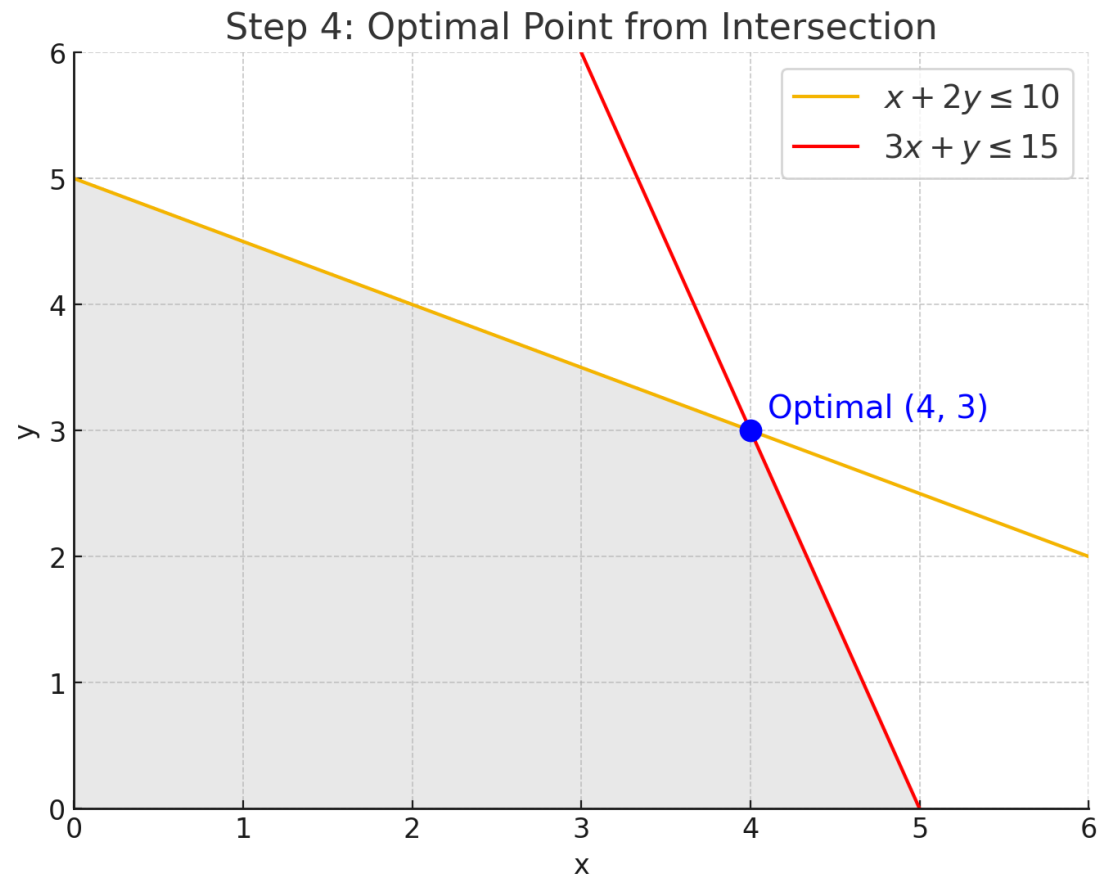
Step 4: Find the coordinates of an optimal point by solving a system of two linear equations from the constraints whose lines pass through this point.

- Solve the system:

$$x + 2y = 10$$

$$3x + y = 15$$

$$\text{Solution} \Rightarrow (x, y) = (4, 3)$$



Graphical Solution Method

Let's look at this LP:

$$\text{Maximize}_{x_1, x_2} 3x_1 + 2x_2$$

s. t.

$$x_1 + x_2 \leq 80$$

$$2x_1 + x_2 \leq 100$$

$$x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

Graphical Solution Method

Try and draw it:

$$\text{Maximize}_{x_1, x_2} 3x_1 + 2x_2$$

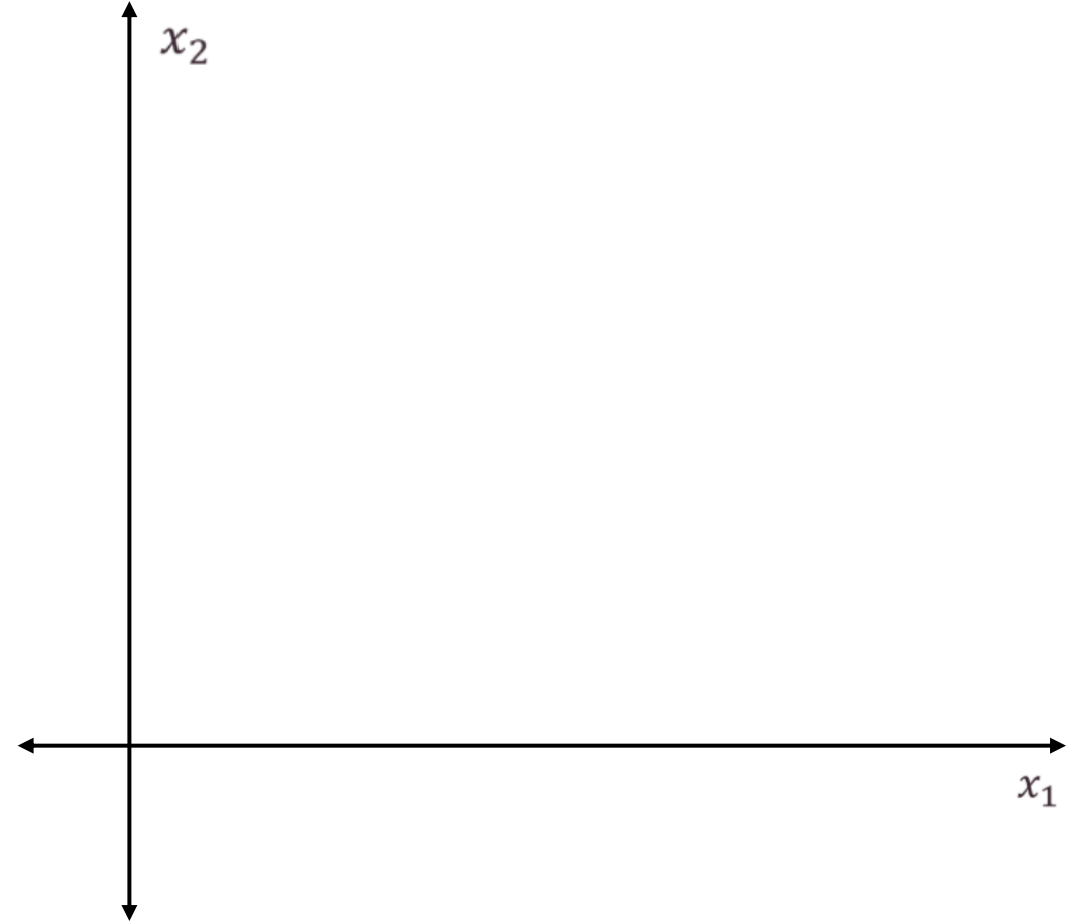
s. t.

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$$x_1 \leq 40$$

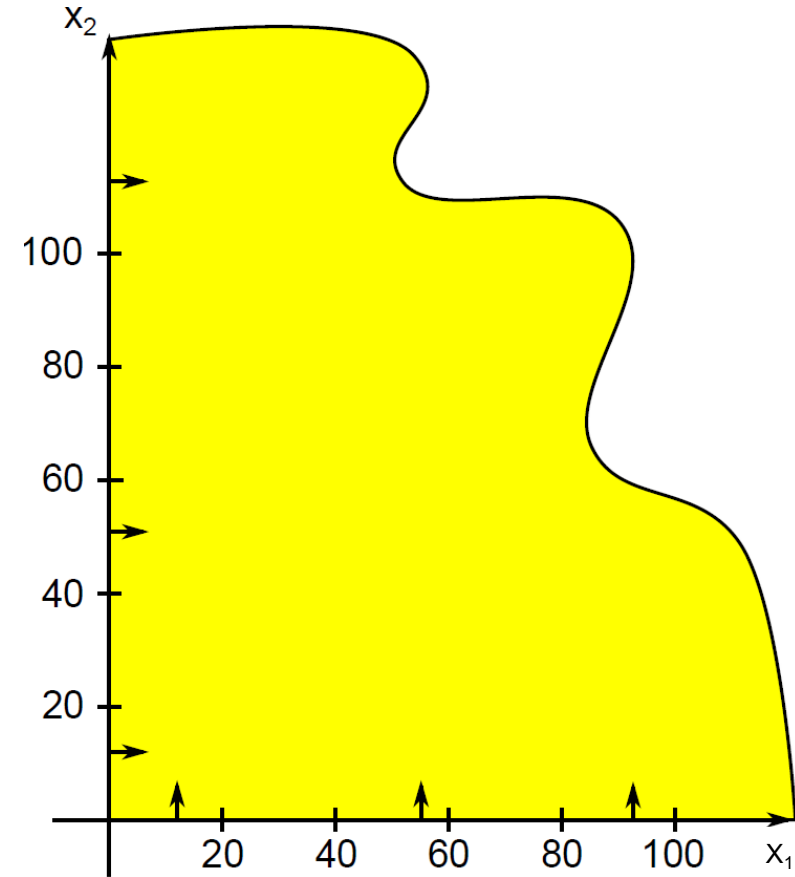
$$x_1, x_2 \geq 0$$



Graphical Solution Method

$$\begin{aligned} & \text{Maximize}_{x_1, x_2} 3x_1 + 2x_2 \\ & \text{s.t.} \end{aligned}$$

$$x_1, x_2 \geq 0$$



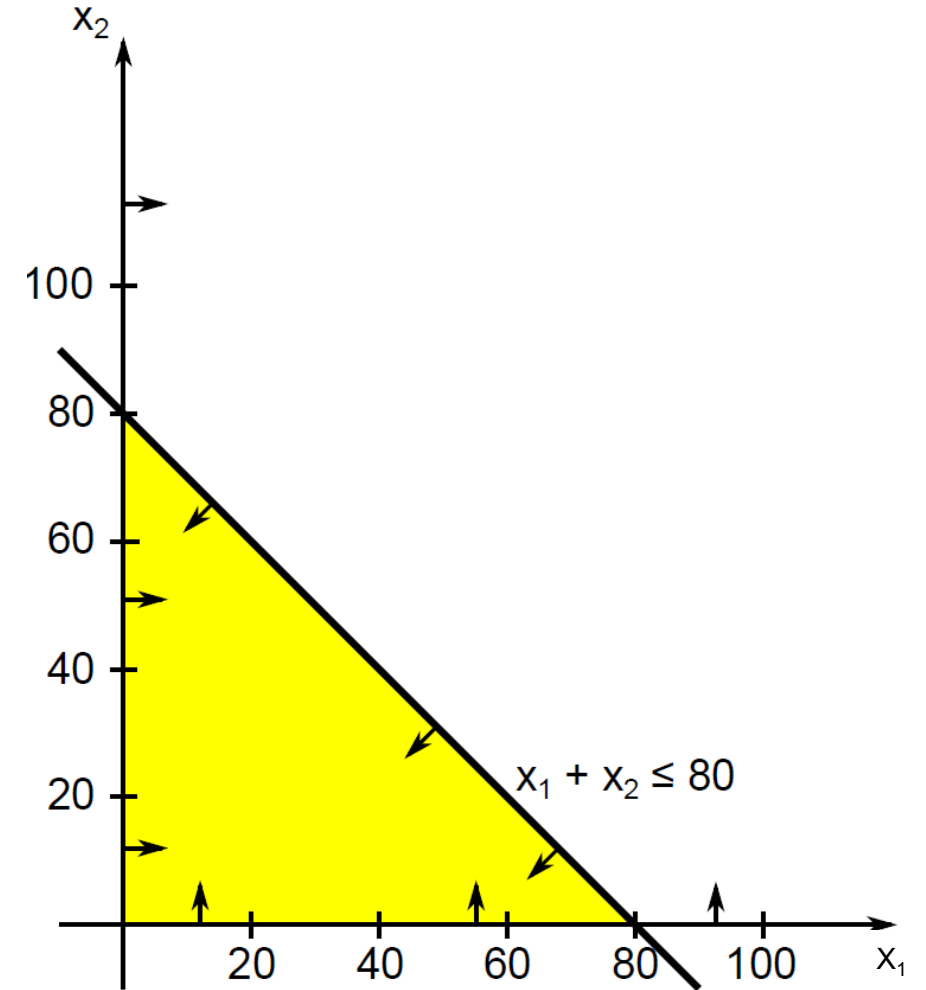
Graphical Solution Method

Maximize _{x_1, x_2} $3x_1 + 2x_2$

s. t.

$$x_1 + x_2 \leq 80$$

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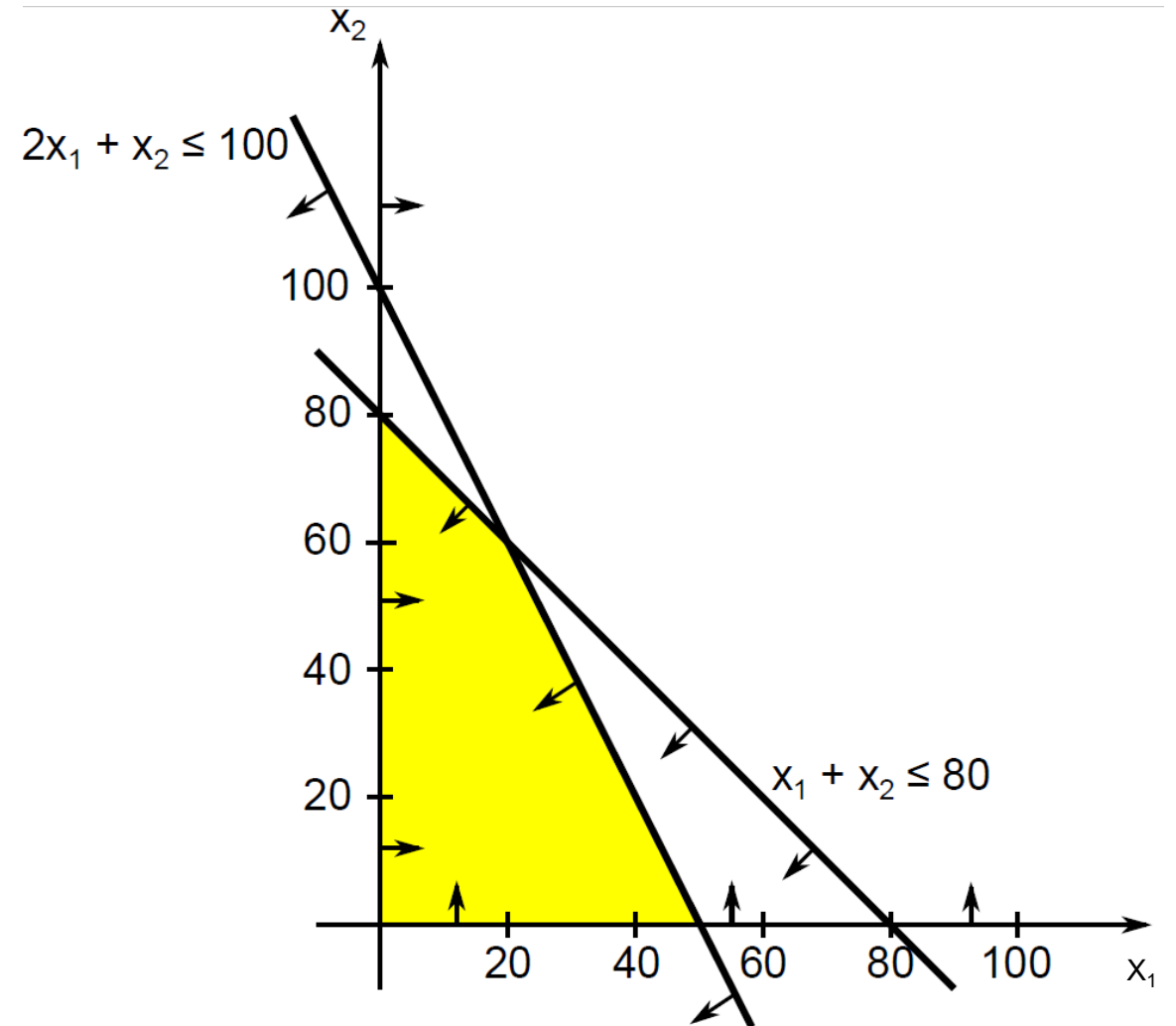
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Graphical Solution Method

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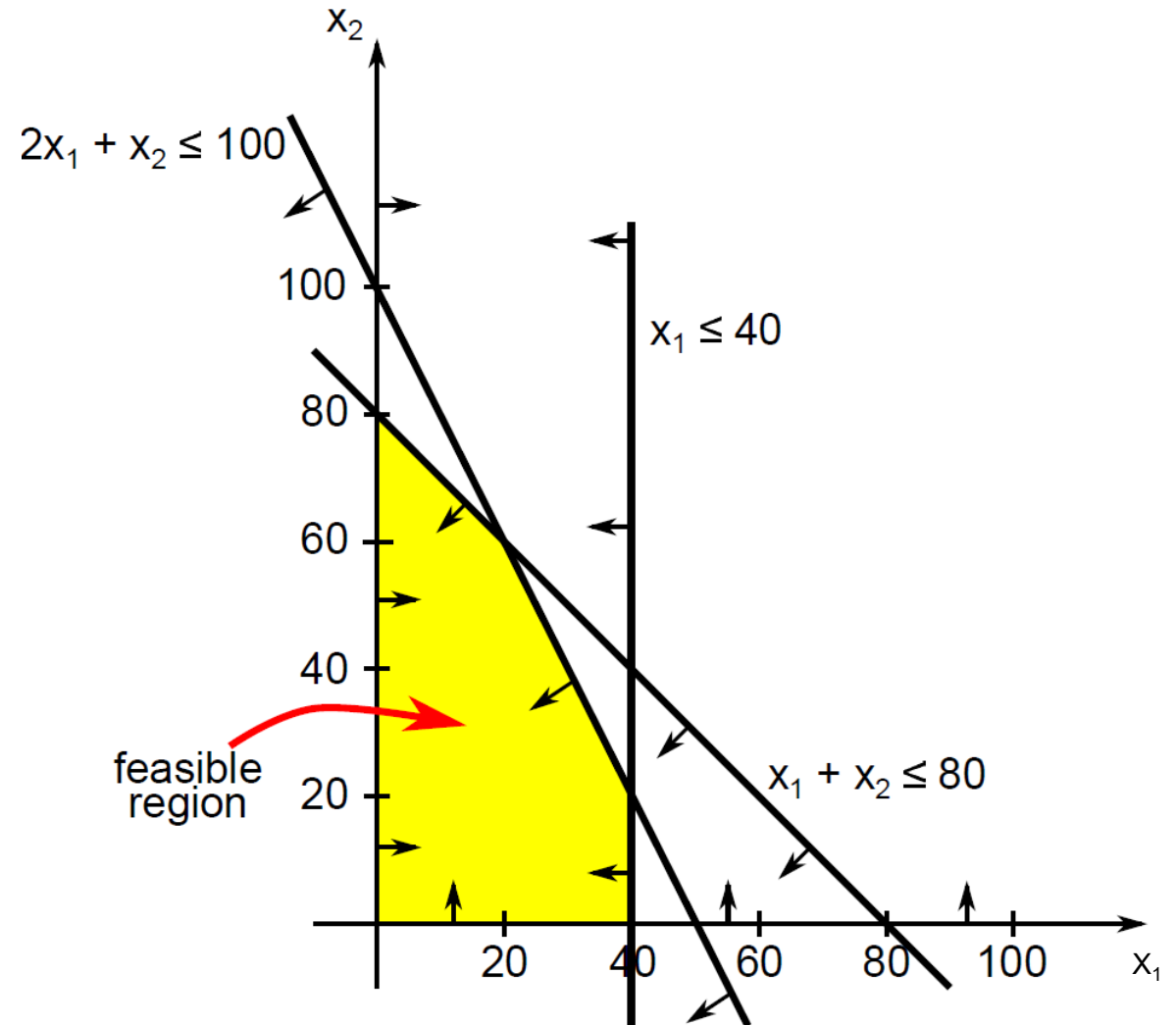
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$$x_1 + x_2 \leq 80$$

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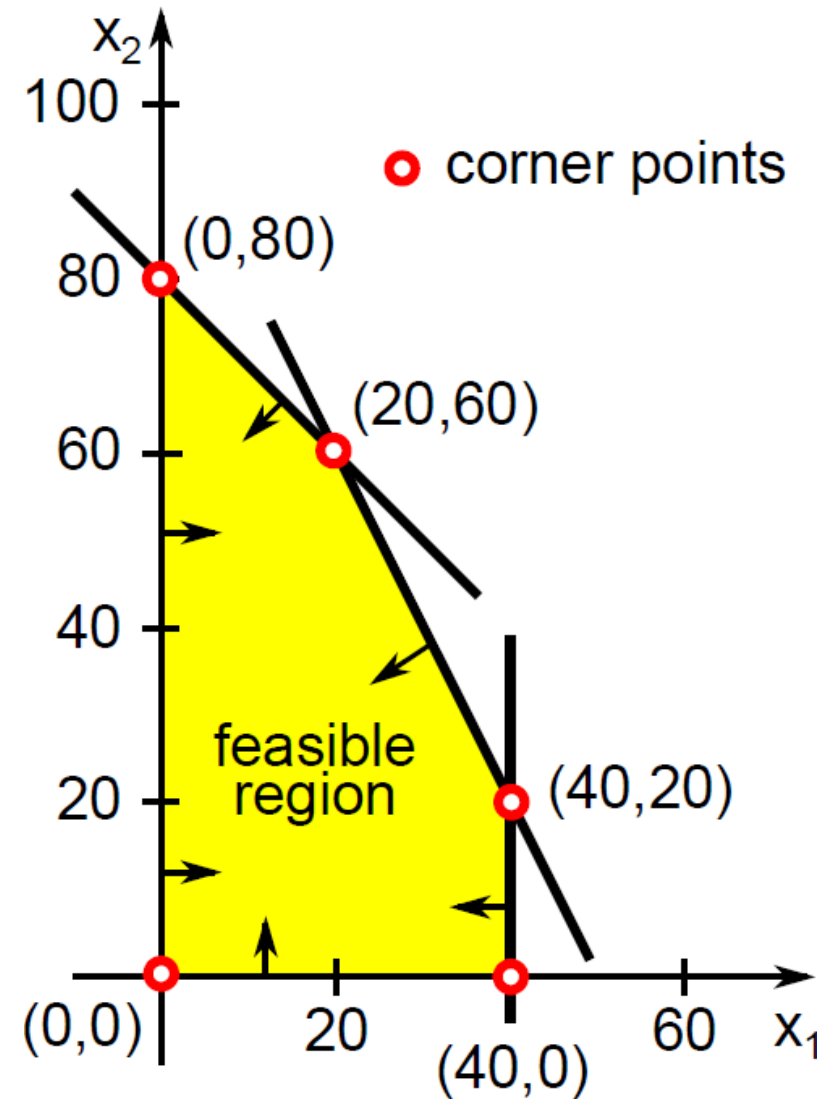
$$x_1 \leq 40$$

$$x_1, x_2 \geq 0$$



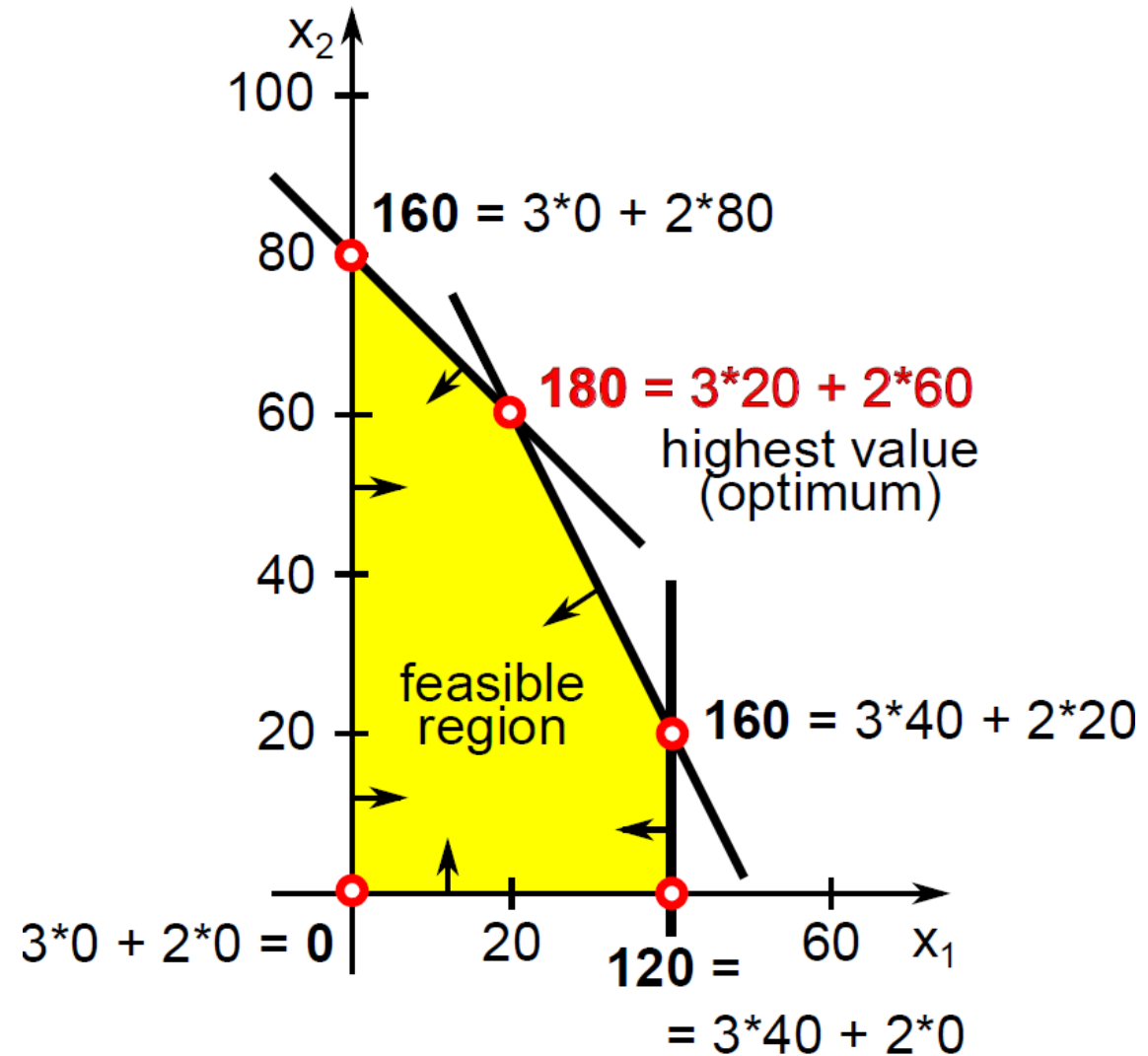
Fun? Fact

If an LP has an optimal solution, it lies at one of the corner point of the feasible region!



Optimal Solution

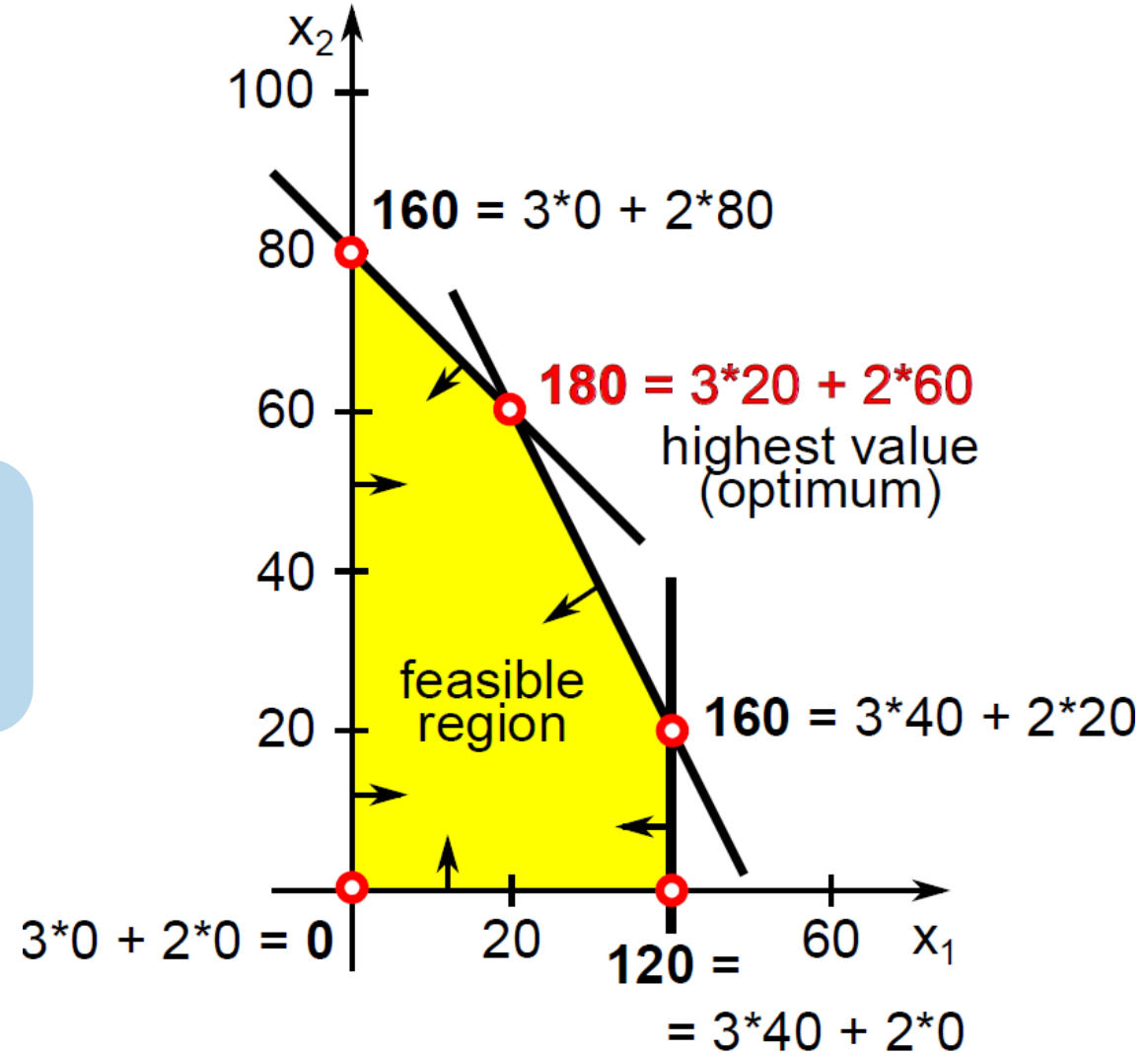
One can check all the corner points...



Optimal Solution

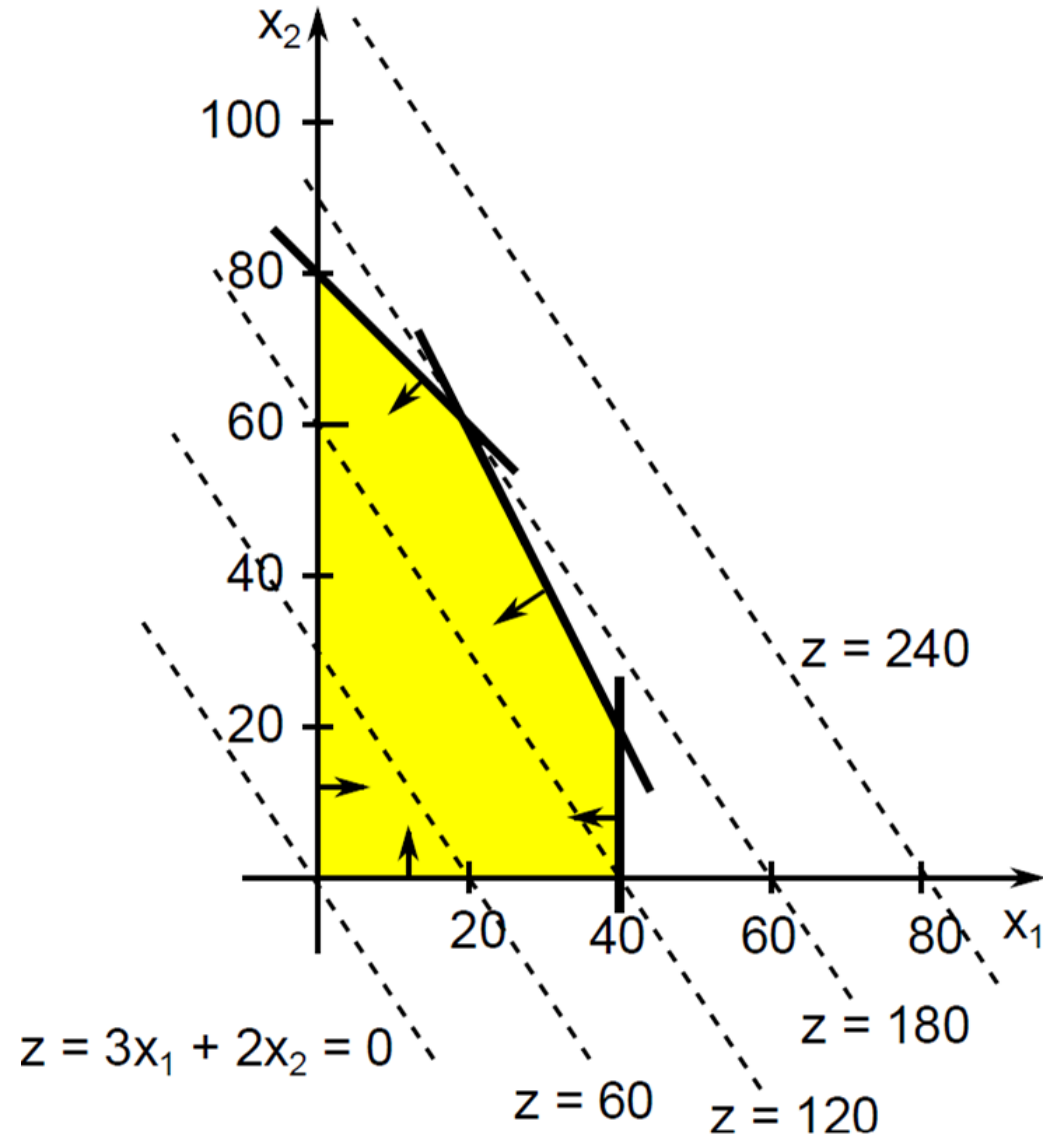
One can check all the corner points...

There may be too many corner points!



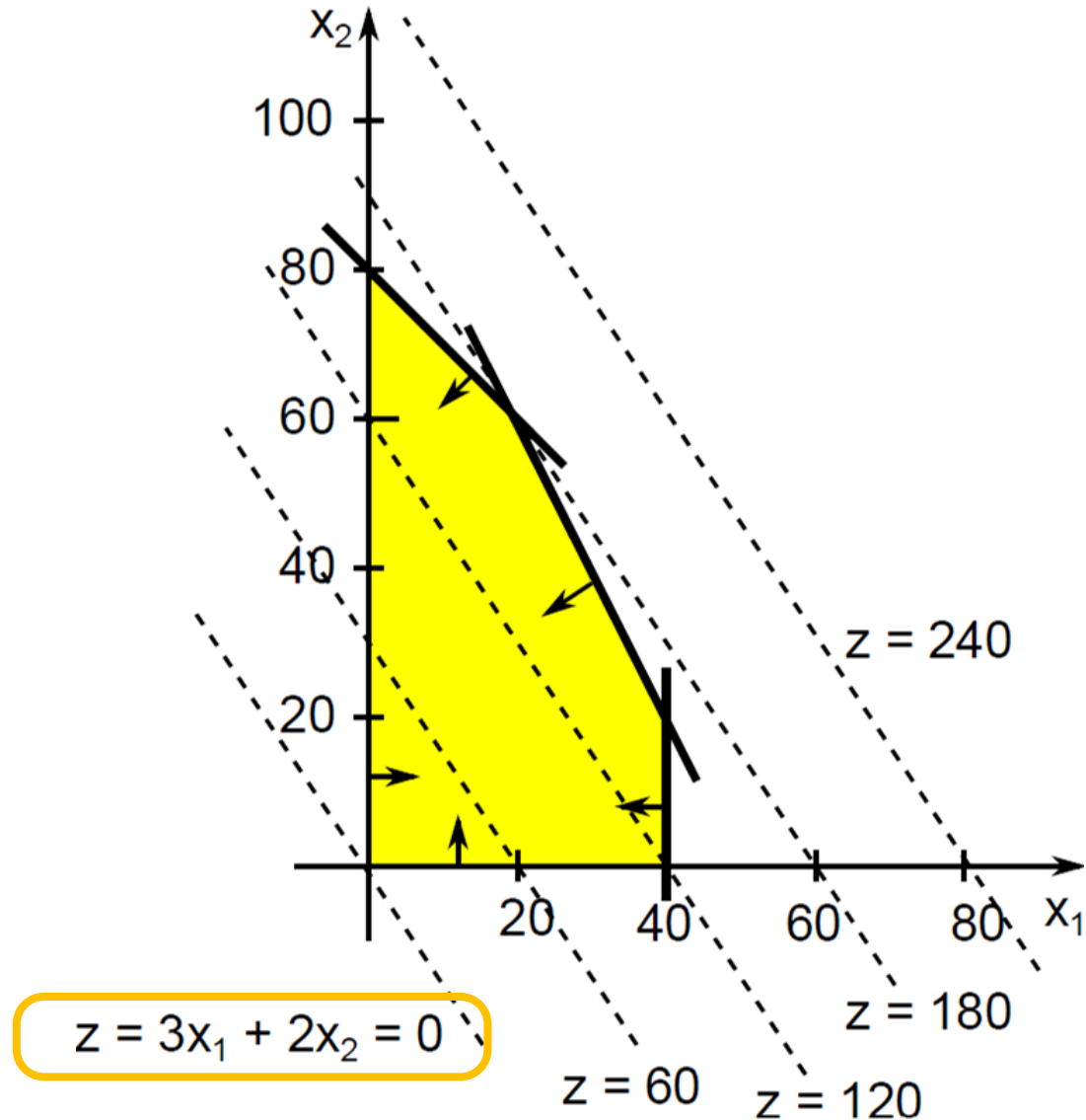
Iso-value Line

We can use the iso-cost
(iso-value) line to find the
optimal solution



Iso-value Line

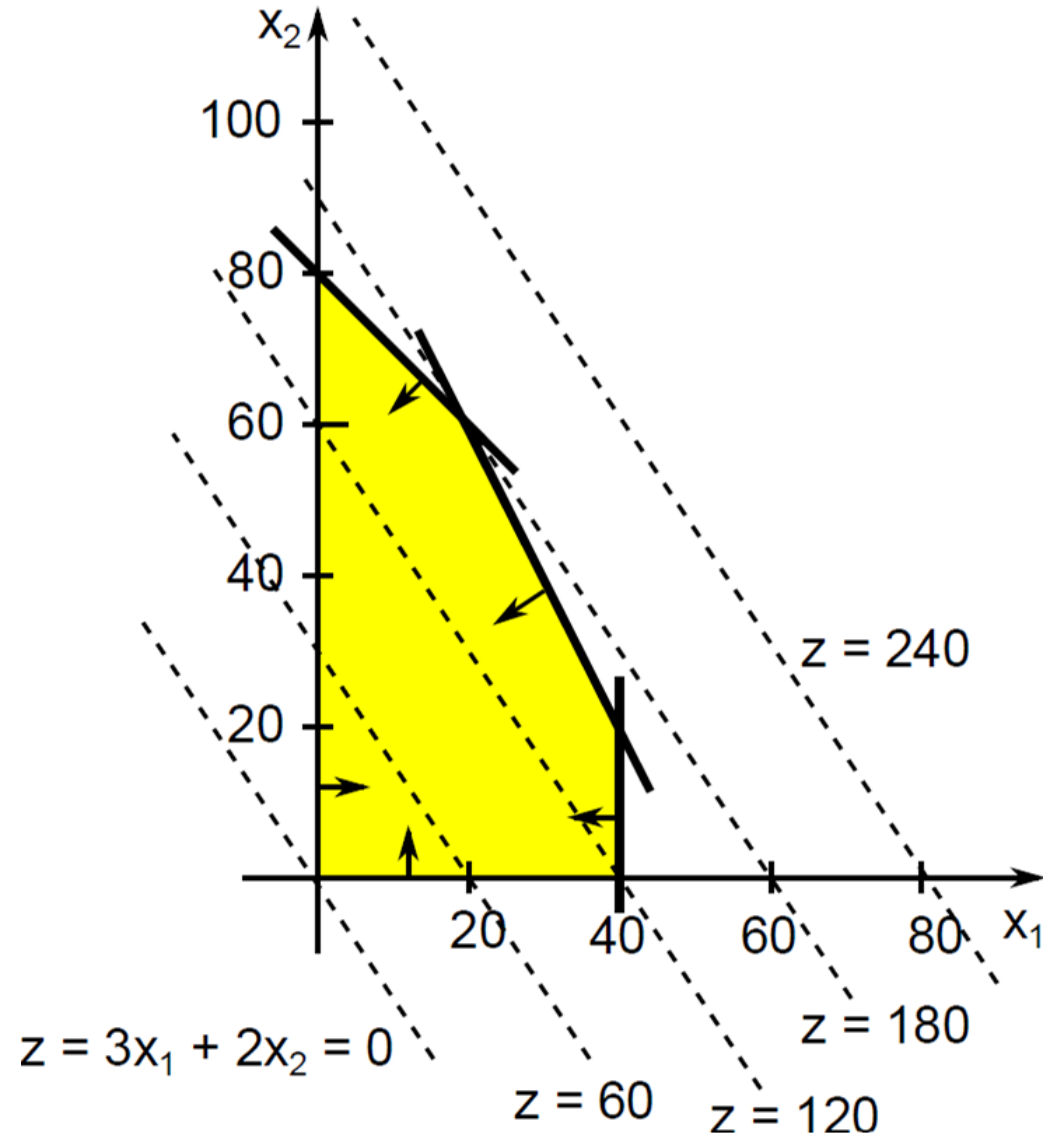
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Iso-value Line

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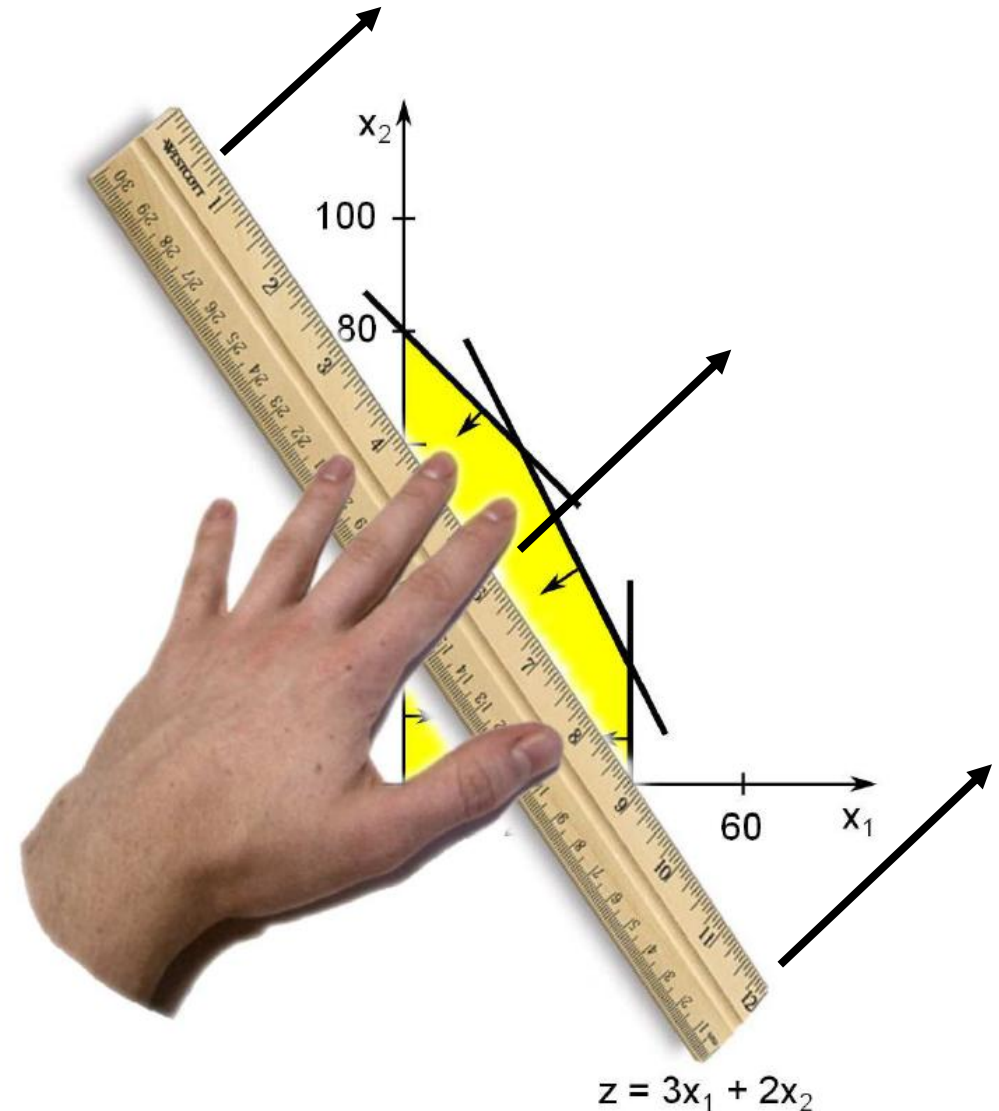
Slide the line in the
increasing direction for
maximization!



Graphical Solution

We can use the iso-cost (iso-value) line to find the optimal solution

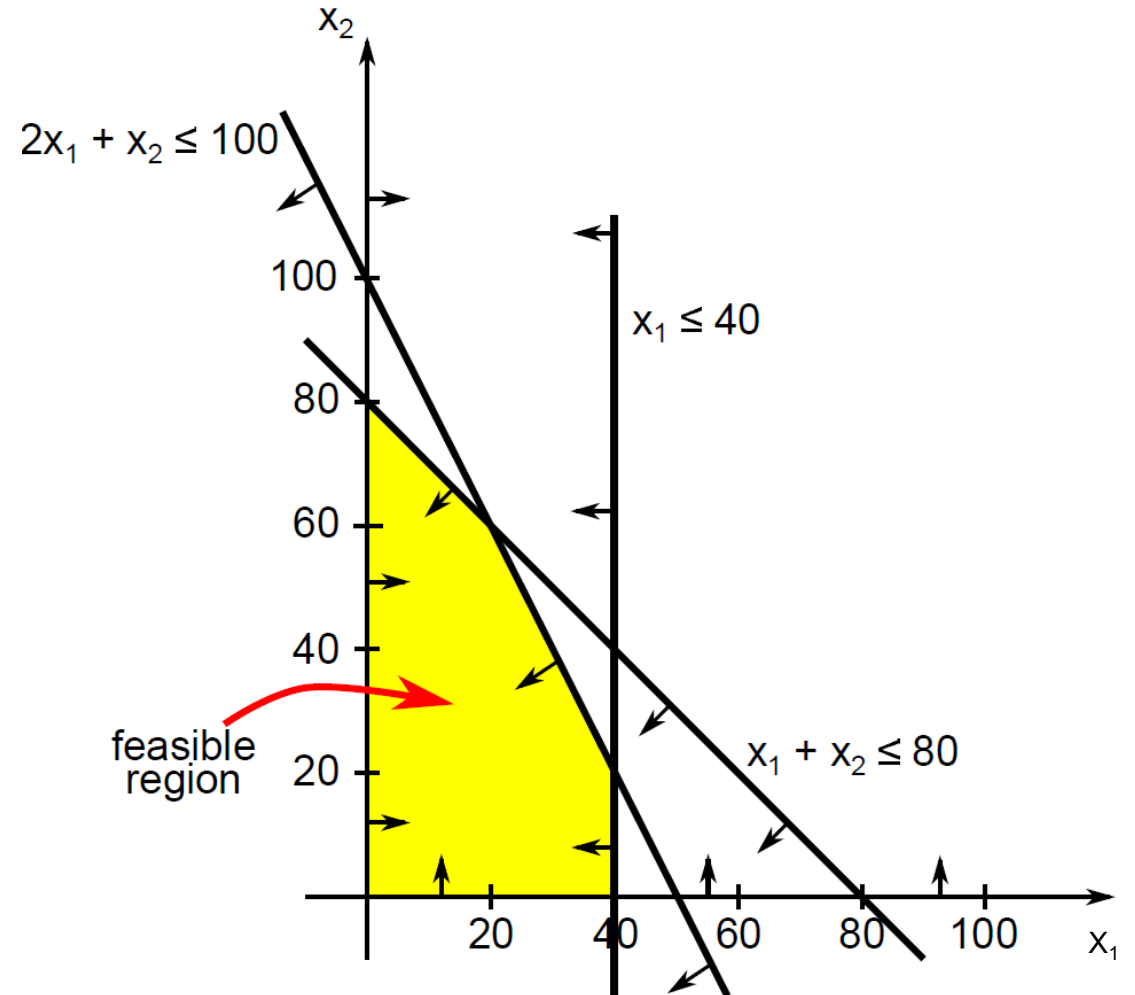
Slide the line in the increasing direction for maximization!



Multiple Optimal Solutions

Suppose the feasibility set is:

$$\begin{cases} x_1 + x_2 \leq 80, & 2x_1 + x_2 \leq 100, \\ x_1 \leq 40, & x_1, x_2 \geq 0 \end{cases}$$



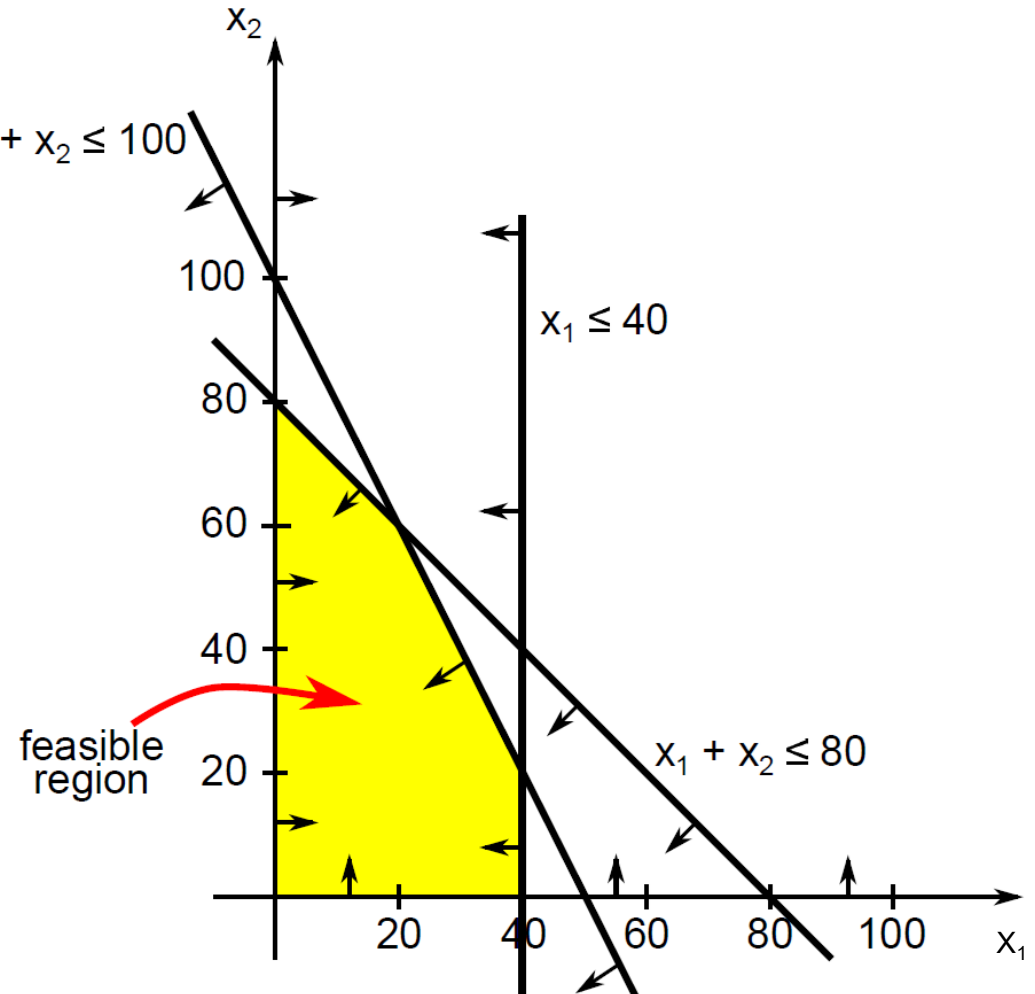
Multiple Optimal Solutions

Suppose the feasibility set is:

$$\{x_1 + x_2 \leq 80, \quad 2x_1 + x_2 \leq 100, \quad x_1 \leq 40, \quad x_1, x_2 \geq 0\}$$

Consider objective function:
 $\max x_1 + x_2$

and try using the iso-value line!



Multiple Optimal Solutions

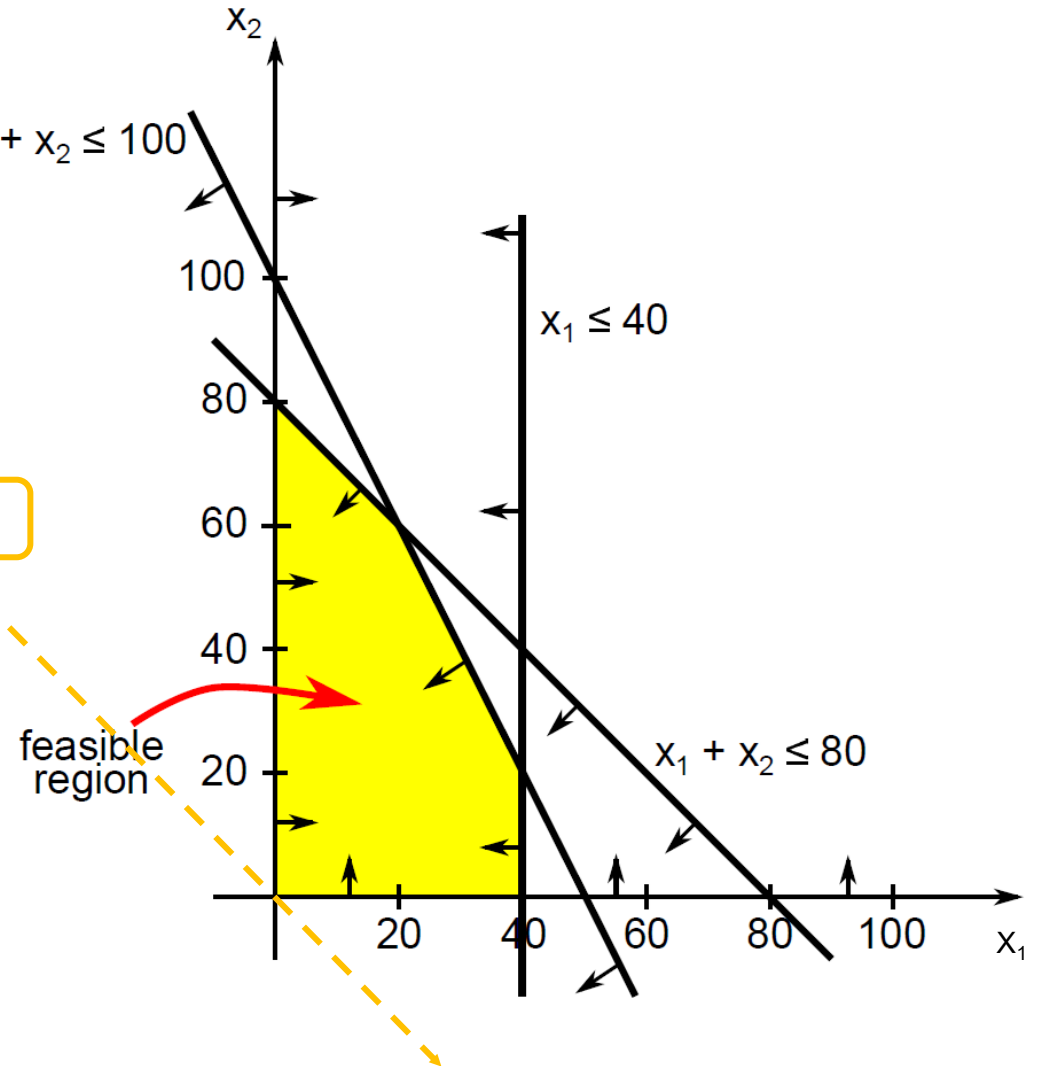
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and try using the iso-value line!

$$x_1 + x_2 = 0$$



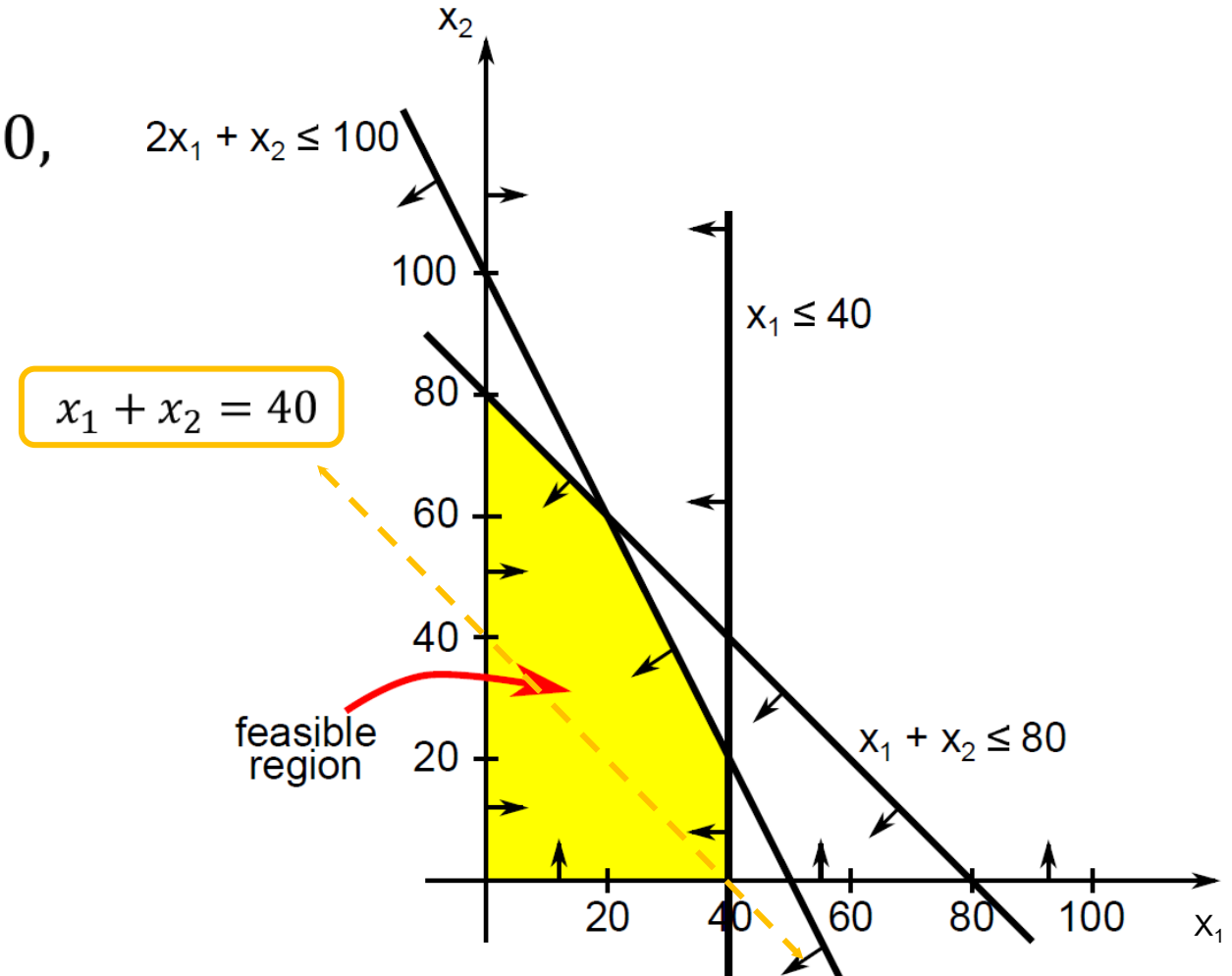
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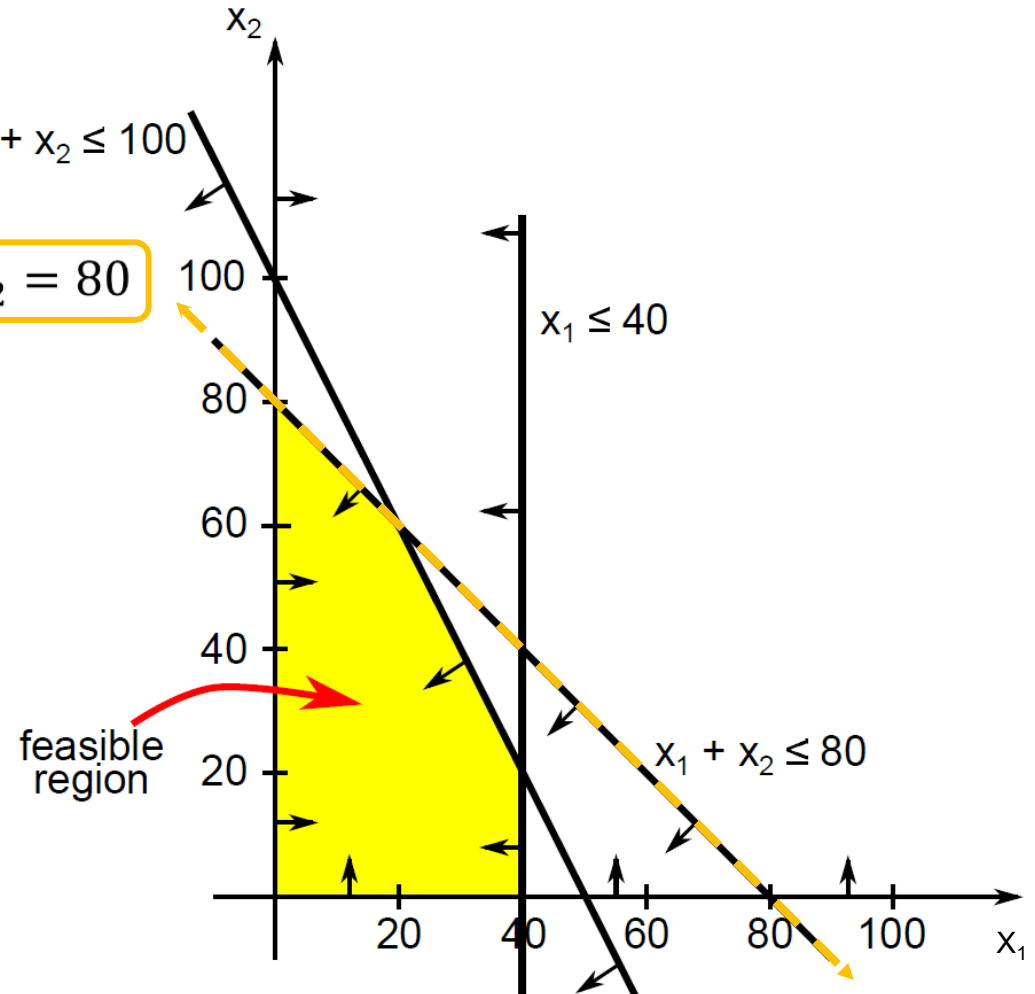
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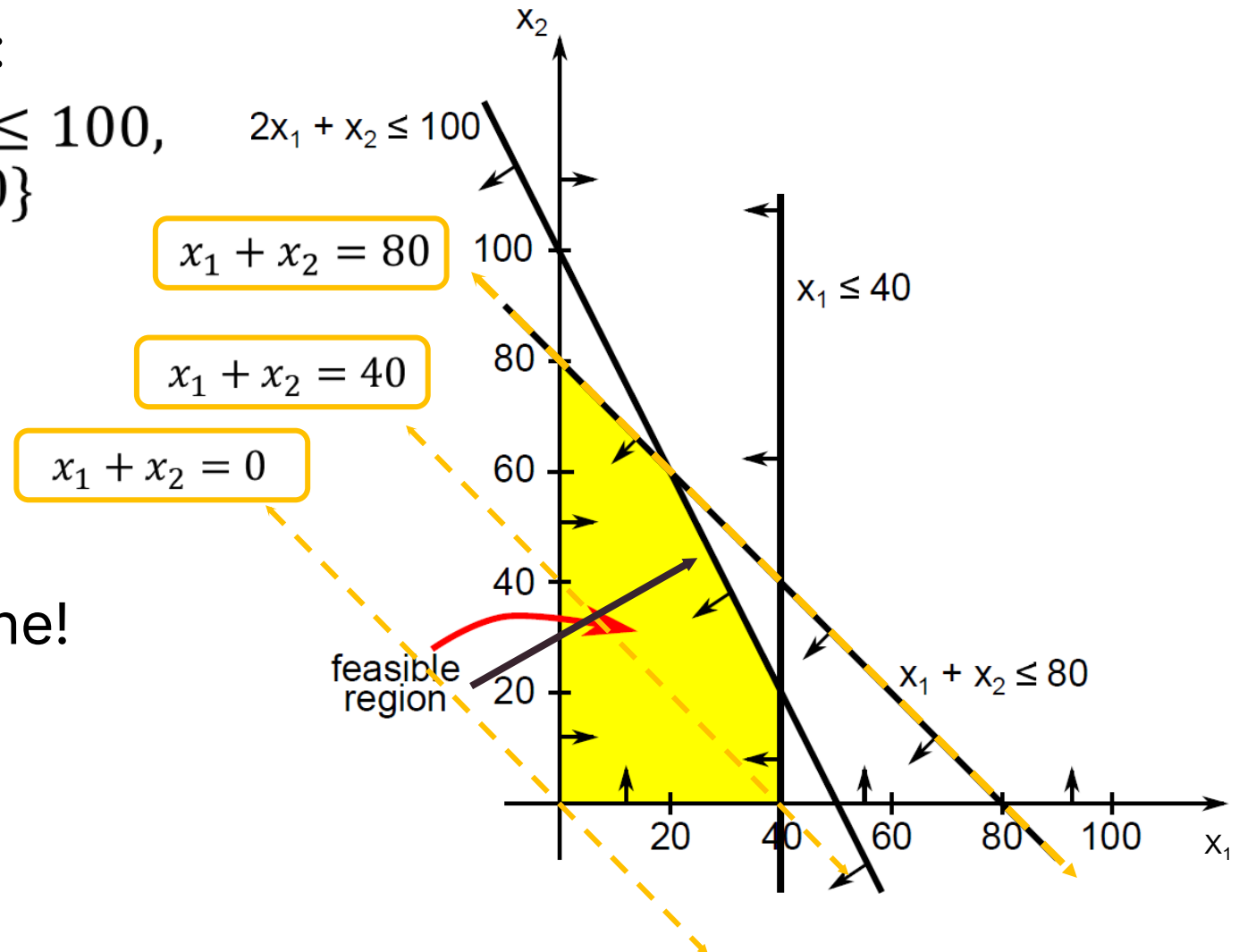
Multiple Optimal Solutions

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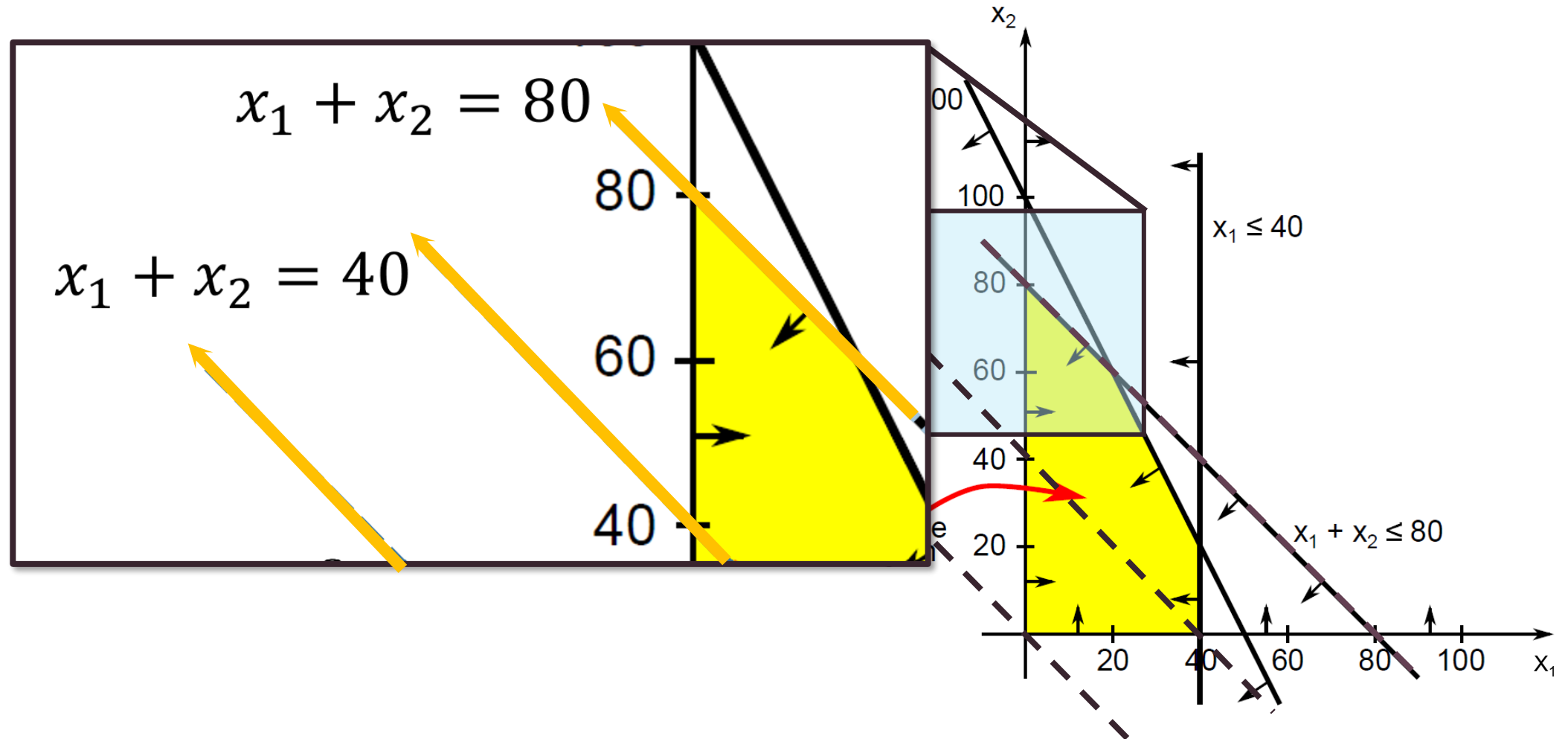
$$\{x_1 + x_2 \leq 80, \quad 2x_1 + x_2 \leq 100, \quad x_1 \leq 40, \quad x_1, x_2 \geq 0\}$$

Consider objective function:
 $\max x_1 + x_2$

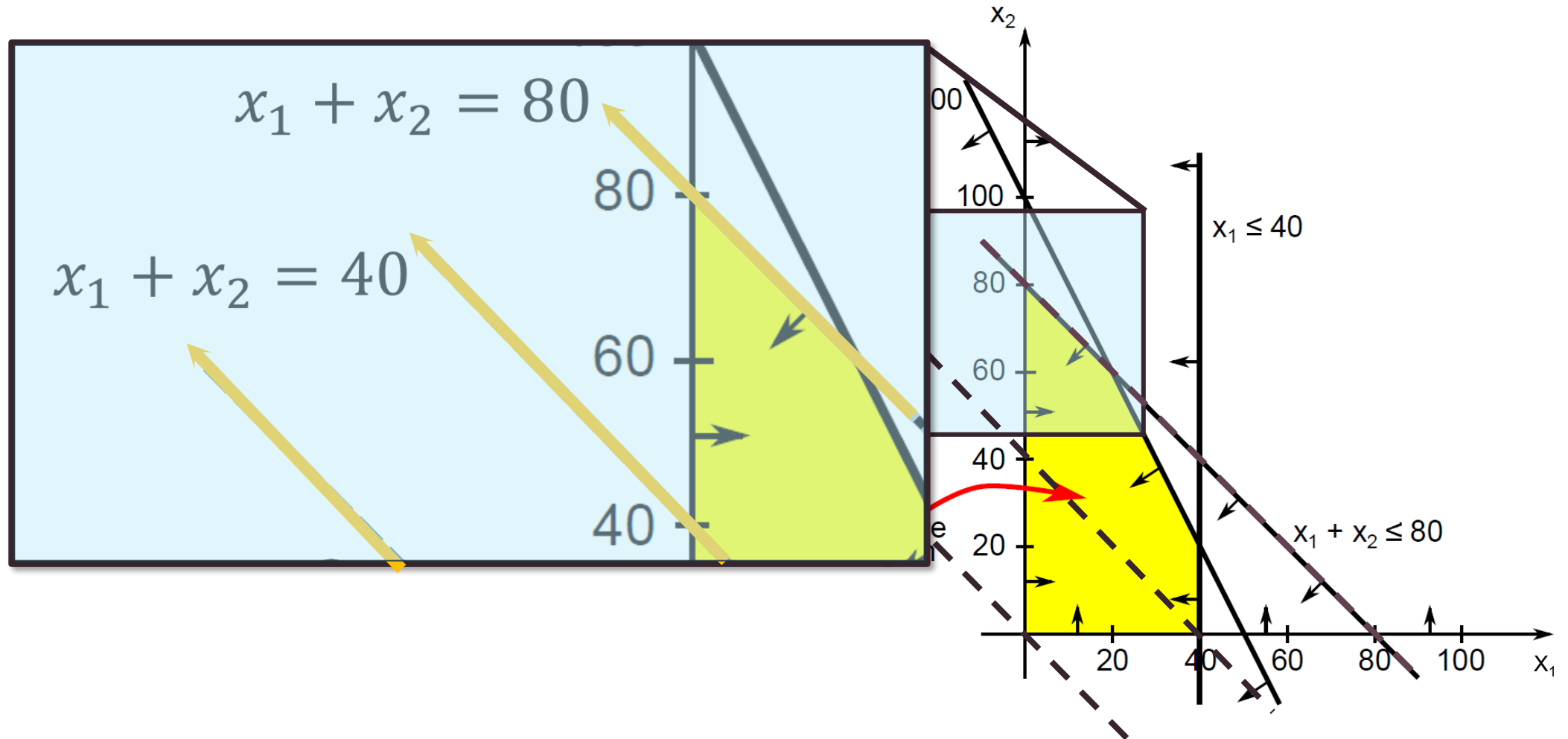
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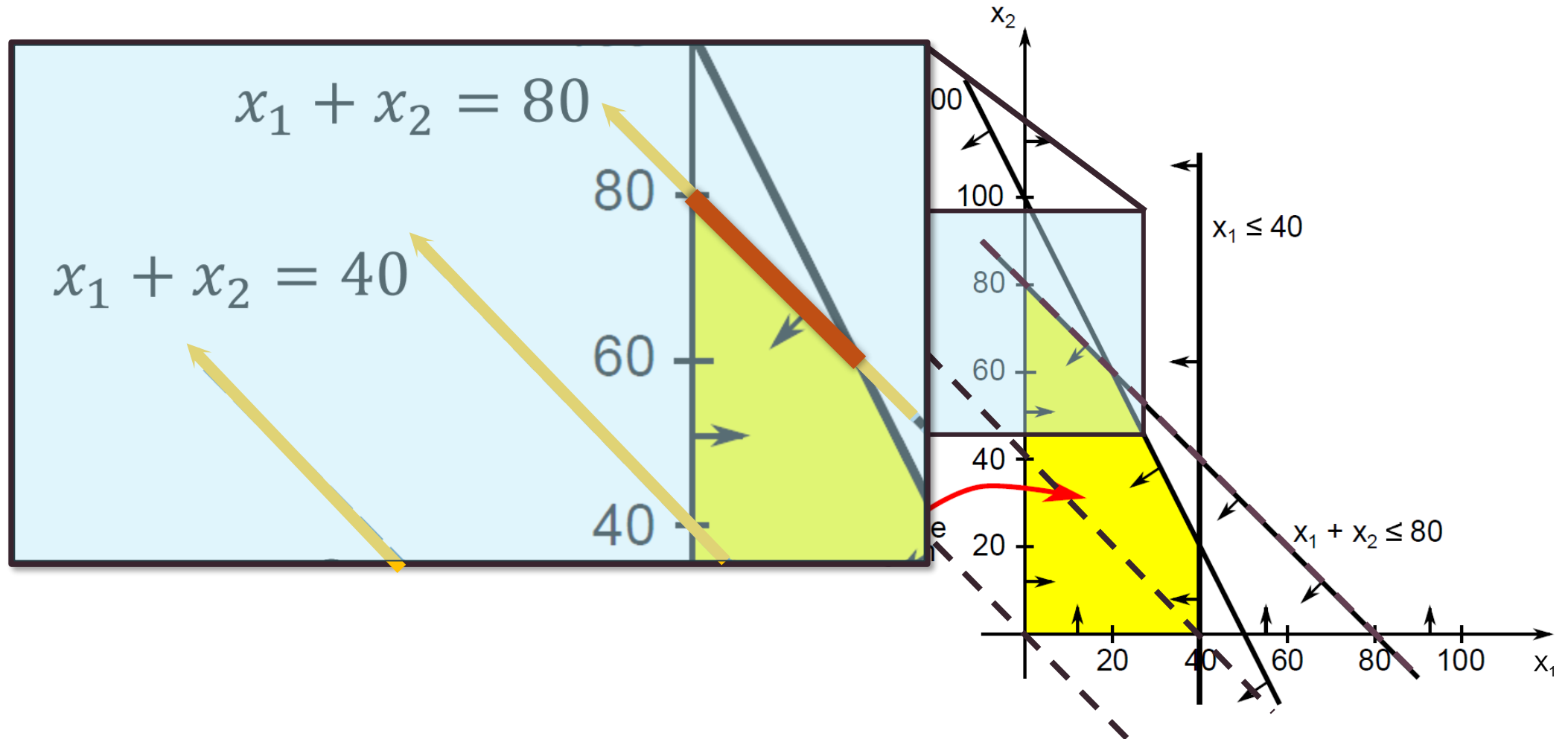
Multiple Optimal Solutions



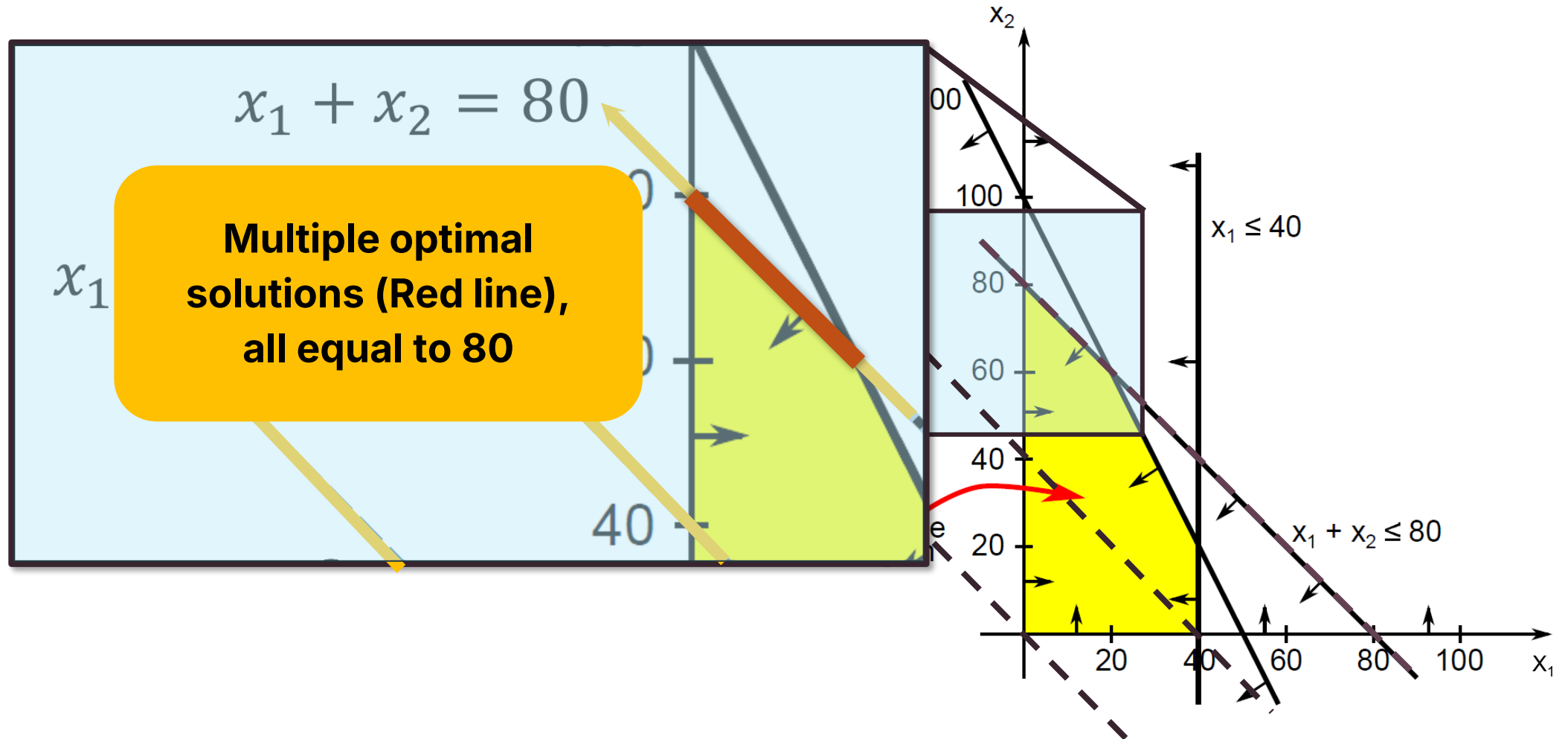
Multiple Optimal Solutions



Multiple Optimal Solutions



Multiple Optimal Solutions



Example | Graphical Solution Method

Let's try to solve this LP:

$$\min z = 50x_1 + 100x_2$$

$$\text{s.t.} \quad 7x_1 + 2x_2 \geq 28$$

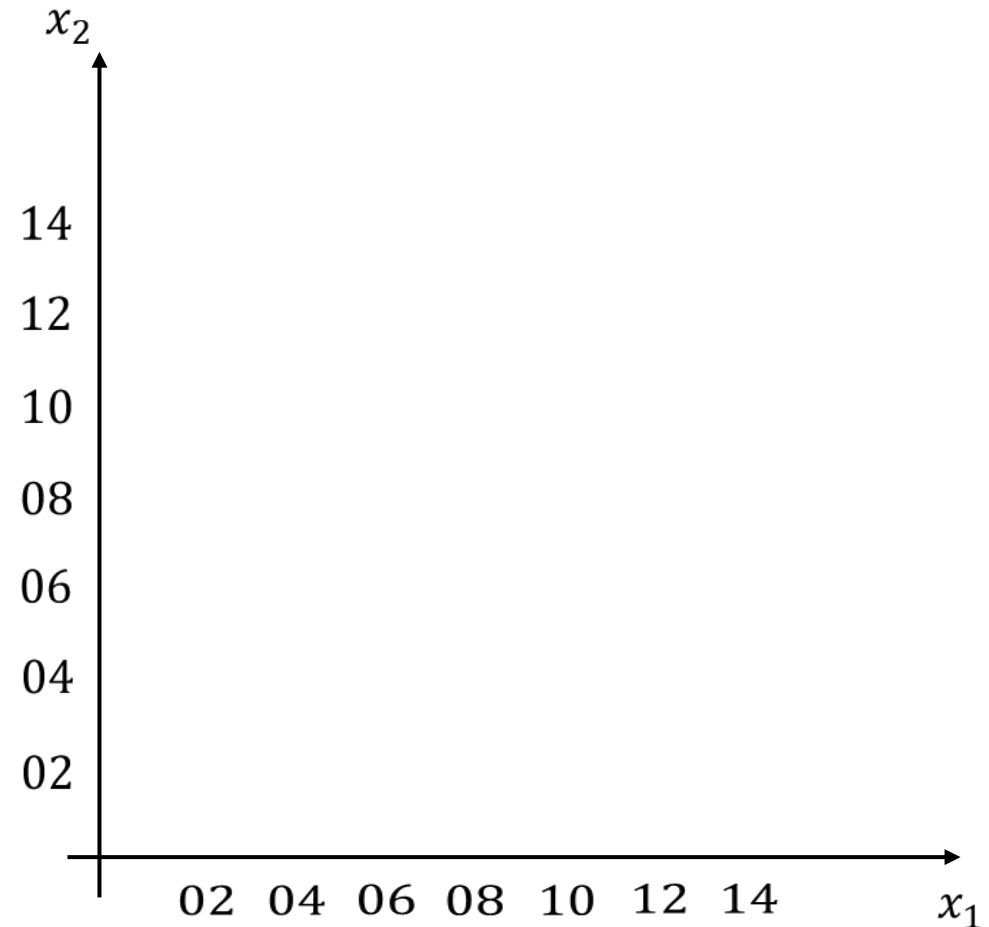
$$2x_1 + 12x_2 \geq 24$$

$$x_1, x_2 \geq 0$$

Example | Graphical Solution Method

Let's try to solve this LP:

$$\begin{array}{ll}\min z = & 50x_1 + 100x_2 \\ \text{s.t.} & 7x_1 + 2x_2 \geq 28 \\ & 2x_1 + 12x_2 \geq 24 \\ & x_1, x_2 \geq 0\end{array}$$

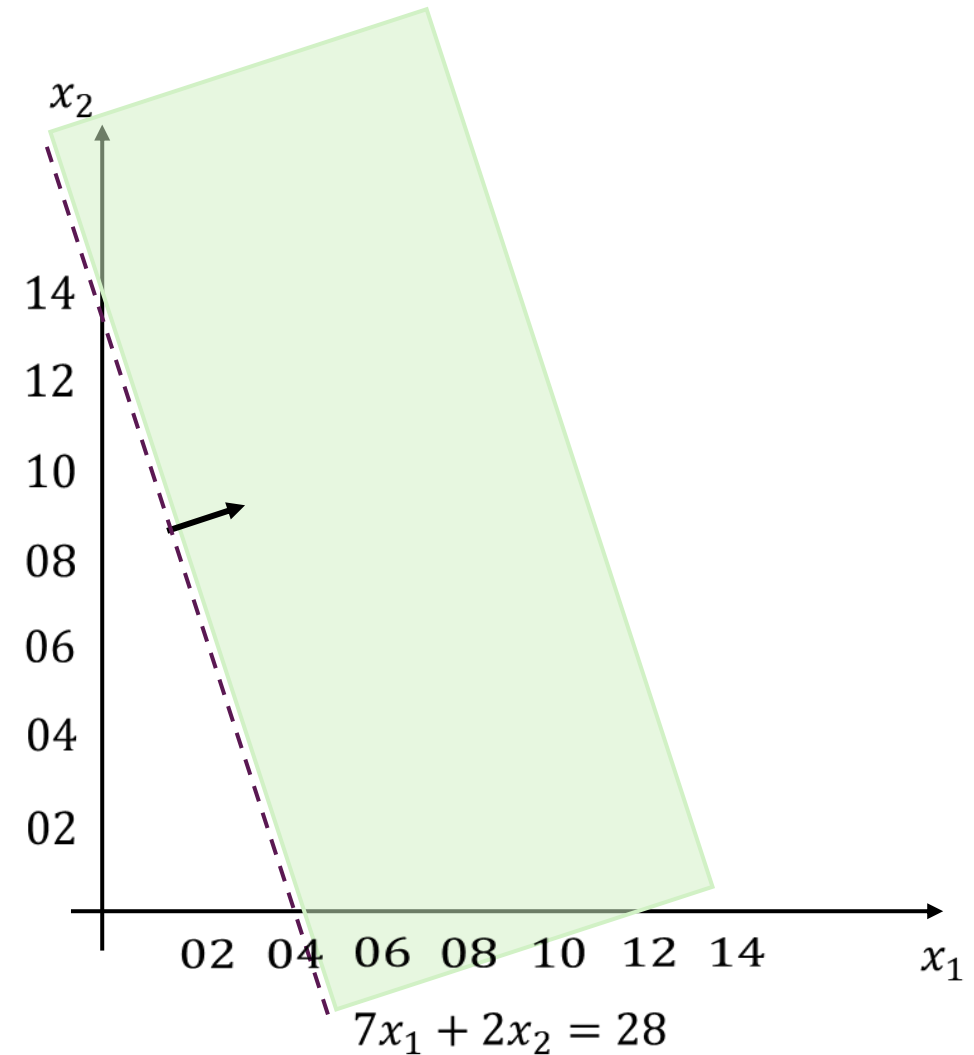


Example | Graphical Solution Method

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$$\begin{array}{ll}\min z = & 50x_1 + 100x_2 \\ \text{s.t.} & 7x_1 + 2x_2 \geq 28 \\ & 2x_1 + 12x_2 \geq 24 \\ & x_1, x_2 \geq 0\end{array}$$

Step 1: Find $7x_1 + 2x_2 \geq 28$



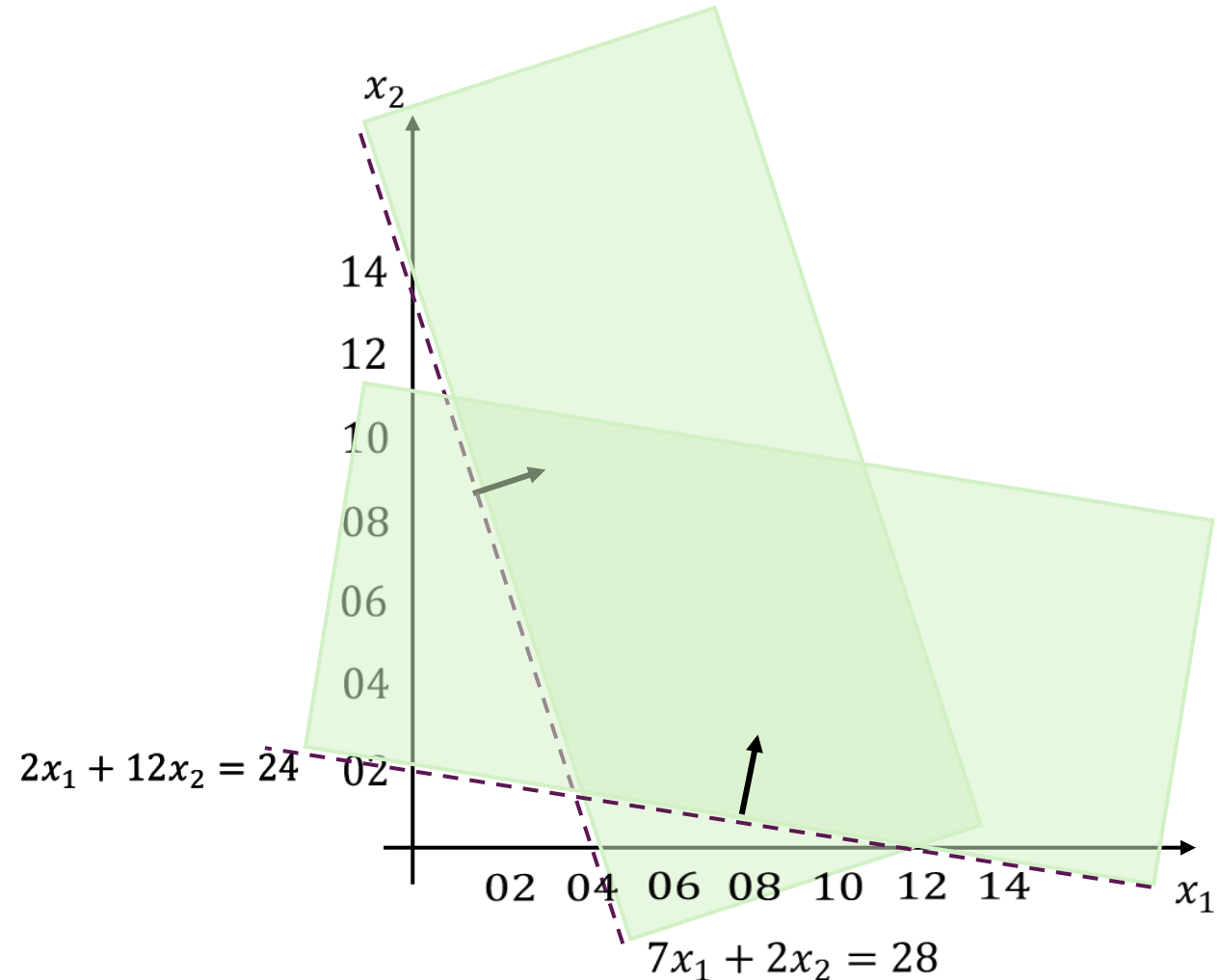
Example | Graphical Solution Method

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Step 1: Find $7x_1 + 2x_2 \geq 28$

Step 2: Find $2x_1 + 12x_2 \geq 24$



Example | Graphical Solution Method

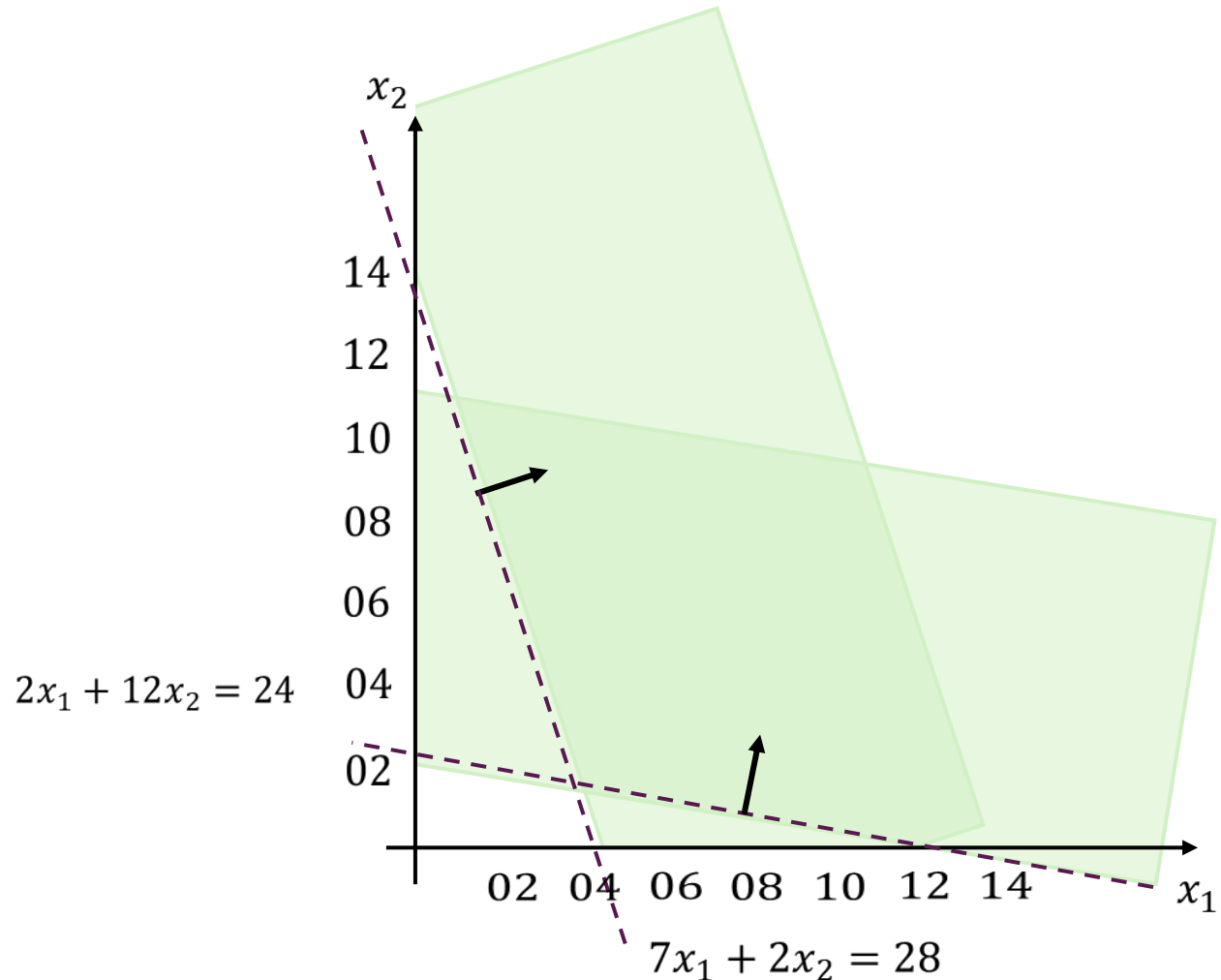
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Step 1: Find $7x_1 + 2x_2 \geq 28$

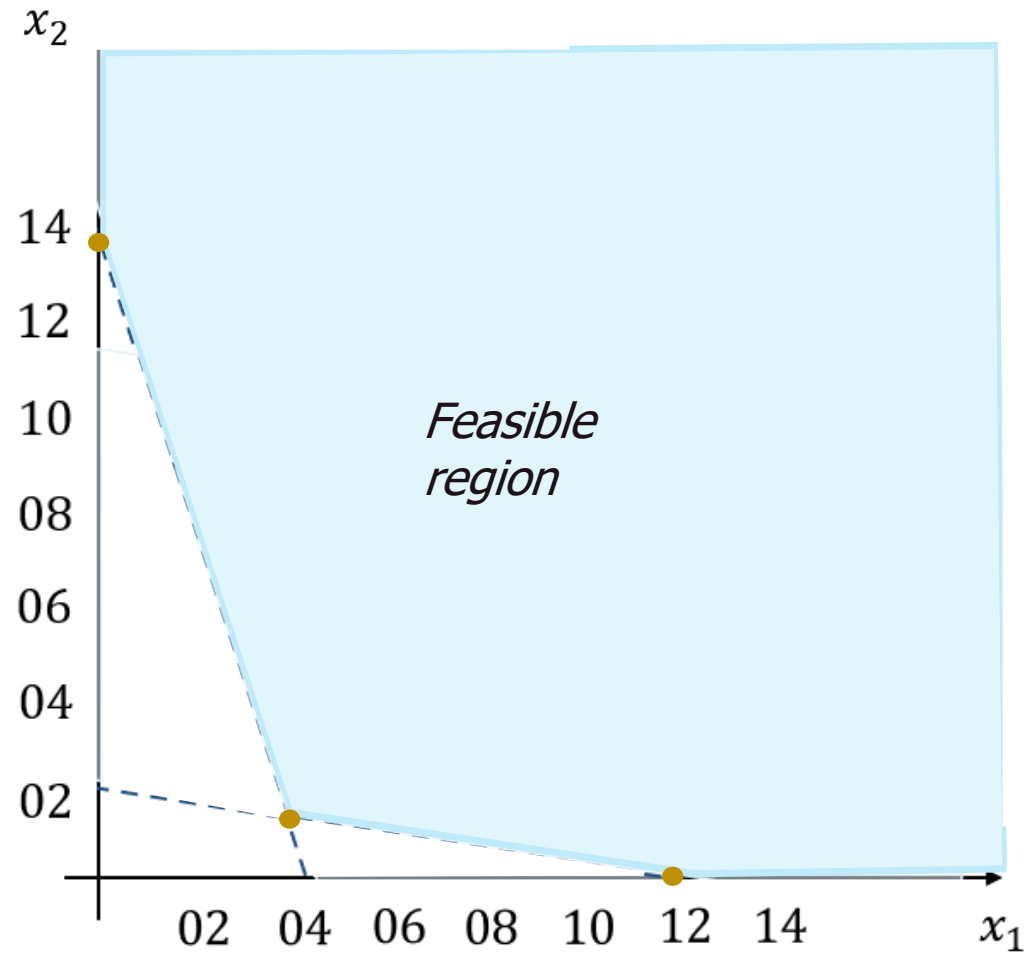
Step 2: Find $2x_1 + 12x_2 \geq 24$

Step 3: Get rid of $x_1, x_2 < 0$



Example | Graphical Solution Method

Step 4: Define the feasible region

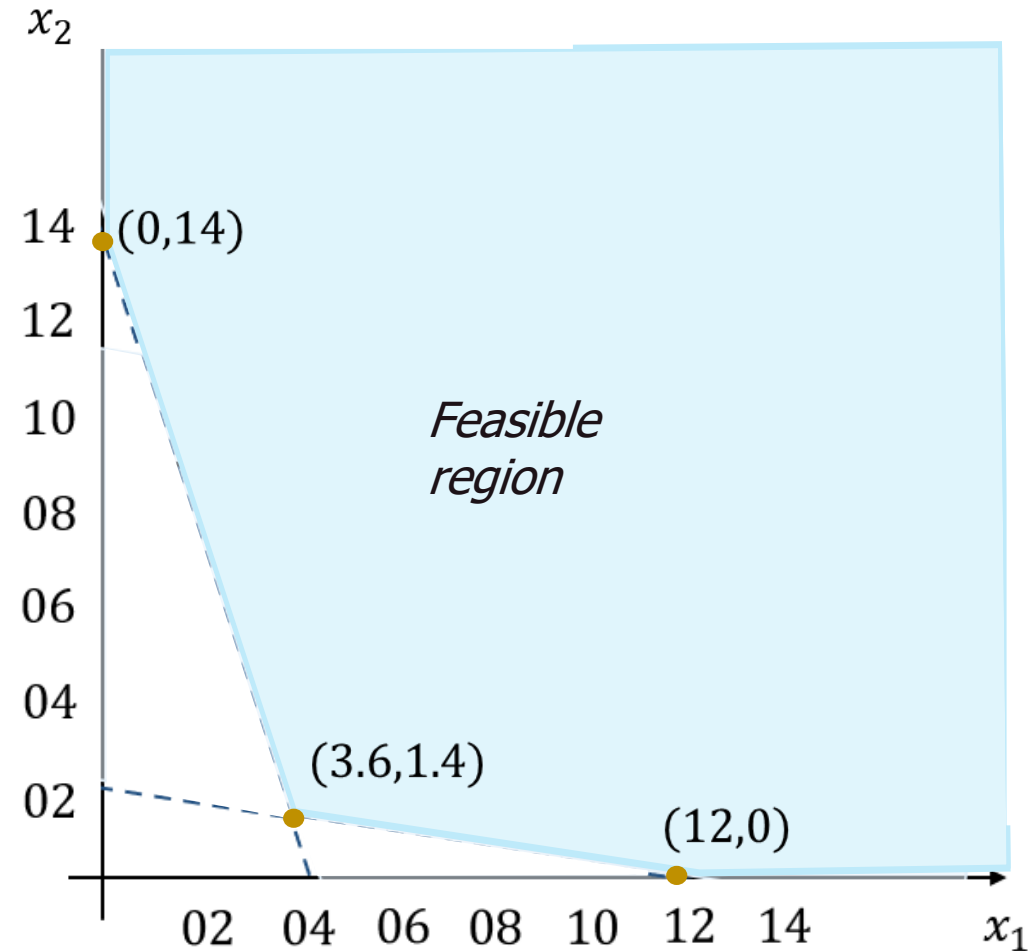


Example | Graphical Solution Method

Step 4: Define the feasible region

Step 5: Find corner points *(if finite)*

$$\left. \begin{array}{l} \text{via } 2x_1 + 12x_2 = 24 \\ \quad 7x_1 + 2x_2 = 28 \end{array} \right\} \text{intersection}$$
$$x_1 = 3.6, x_2 = 1.4$$



Example | Graphical Solution Method

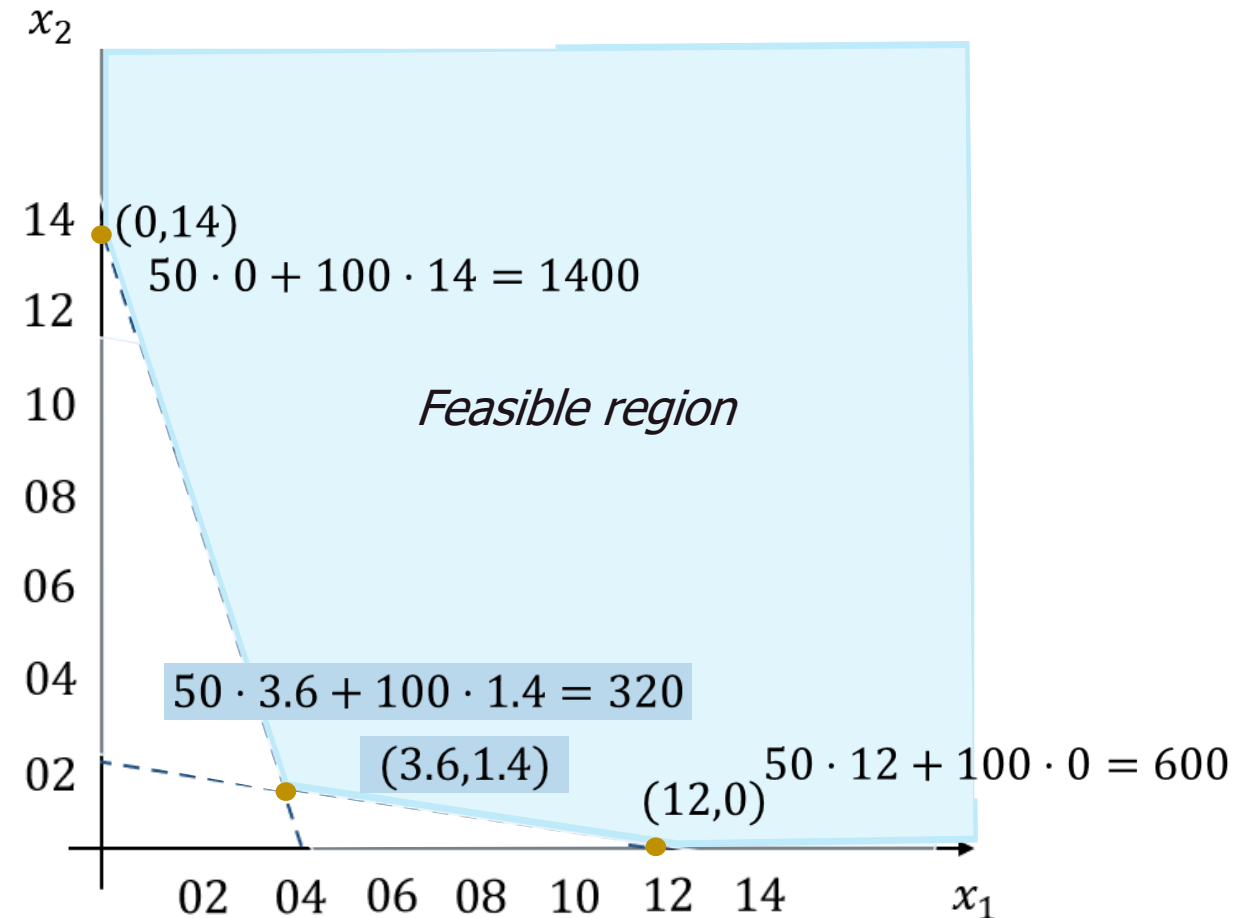
Step 4: Define the feasible region

Step 5: Find corner points *(if finite)*

$$\left. \begin{array}{l} \text{via } 2x_1 + 12x_2 = 24 \\ 7x_1 + 2x_2 = 28 \end{array} \right\} \text{intersection}$$
$$x_1 = 3.6, x_2 = 1.4$$

Step 6: Compare corner point values and pick the optimal!

Note: Objective function is to minimize z and not maximize!

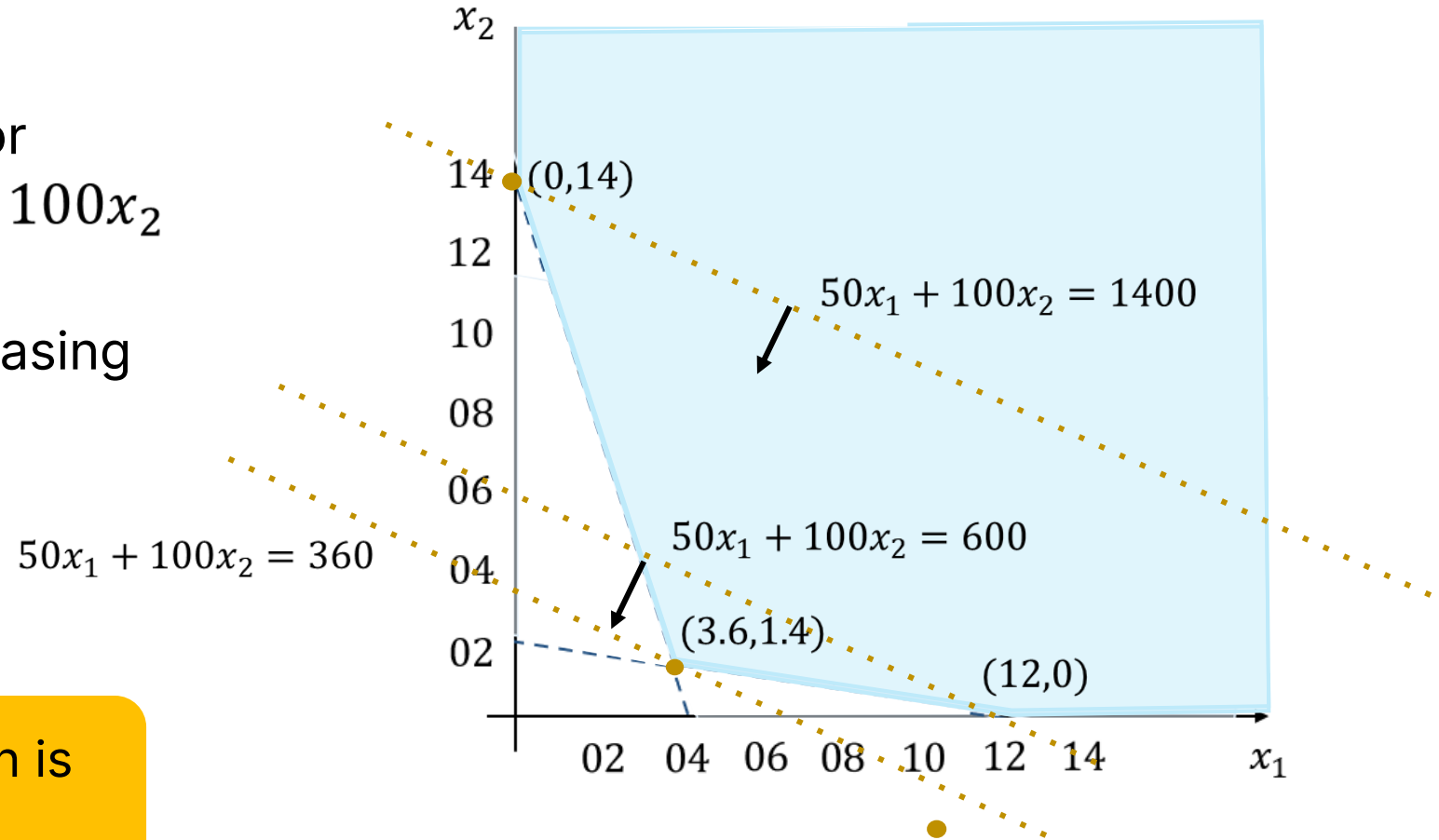


Example | Graphical Solution Method

Different approach:

- Draw iso-value line for
 $\min z = 50x_1 + 100x_2$
- Slide the line in decreasing direction (minimization problem)

The optimal solution is
 $x_1, x_2 = (3.6, 1.4)$



Reading Suggestion:

Extreme Points and Basic Feasible Solutions

Section 7.1 from Veatch's Book, pg 134-136

Introduction to Solvers

Topics

Basic Programming Examples:

- Portfolio Optimization Using Excel Solver
- Vaccine Allocation Problem Using Gurobi



Problem Statement:

Suppose you are a financial engineer advising a client on asset allocation. Your client has \$100,000 and you have narrowed the options down to five different investment vehicles:

- U.S. equities
- Investment-grade corporate bonds
- U.S. Treasuries
- REITs (Real Estate Investment Trusts)
- International equities

Excel | Portfolio Optimization



Asset	Exp. Return (μ_i)	ESG score (e_i)	Liquidity (ℓ_i , 1–10)	Duration (d_i , yrs)	Trans. cost (k_i , %)	Bad year loss (b_i)
U.S. equities	8.0%	6.0	9.0	0.0	0.40%	-15%
Int'l equities	7.0%	7.0	8.0	0.0	0.40%	-12%
Investment- grade corps	5.5%	6.5	8.5	6.0	0.20%	-3%
U.S. Treasuries	4.0%	8.0	10.0	5.0	0.20%	+2%
REITs	6.2%	5.5	7.0	0.0	0.30%	-10%

Probability of having a bad year is 10%

Excel | Portfolio Optimization



The client has set the following requirements:

1. At least 5% of the assets need to remain in cash,
2. At least 30% of their portfolio should be in bonds (corporates + treasuries) to maintain a conservative posture
3. No more than 50% of the portfolio should be in stocks (U.S. + Int'l)
4. At least 10% should be invested in REITs for diversification.

Excel | Portfolio Optimization



Then you add the risk management and safety constraints:

1. At least 20% of the portfolio must be in liquid assets (treasuries + cash)
2. Average ESG score must be at least 7
3. The potential loss in a bad year should not be more than \$8,000



Mathematical Model

Let x_i represent the proportion of the portfolio to be invested in different assets, where x_6 represents the amount left as cash.

Objective Function:

Maximize the expected return of the portfolio while adhering to the client's & risk constraints.

$$\text{Maximize: } \sum_{i=1}^6 (\mu_i - k_i) x_i$$

Excel | Portfolio Optimization



Subject to constraints:

$$\sum_{i=1}^6 x_i = 1$$

(The sum of the proportions must be 1)

$$x_6 \geq 0.05$$

(At least 5% in cash)

$$x_5 \geq 0.1$$

(At least 10% in real estate)

$$x_3 + x_4 \geq 0.3$$

(At least 30% in bonds)

$$x_1 + x_2 \leq 0.5$$

(At most 50% in stocks)

Excel | Portfolio Optimization



Subject to constraints cont'd:

$$\frac{\sum_{i=1}^5 e_i x_i}{\sum_{i=1}^5 x_i} \geq 7 \quad (\text{Average ESG score must be at least 7})$$

$$x_6 + x_4 \geq 0.2 \quad (\text{Liquidity})$$

$$\sum_{i=1}^6 b_i x_i \geq -8,000 \quad (\text{Safety})$$

Excel | Portfolio Optimization



The complete LP

Maximize:
$$\sum_{i=1}^6 (\mu_i - k_i) x_i$$

Subject to:

$$\sum_{i=1}^6 x_i = 1 \quad (1)$$

$$x_6 \geq 0.05 \quad (2)$$

$$x_5 \geq 0.1 \quad (3)$$

$$x_3 + x_4 \geq 0.3 \quad (4)$$

$$x_1 + x_2 \leq 0.5 \quad (5)$$

$$\frac{\sum_{i=1}^6 e_i x_i}{\sum_{i=1}^6 x_i} \geq 7 \quad (6)$$

$$x_6 + x_4 \geq 0.2 \quad (7)$$

$$\sum_{i=1}^6 b_i x_i * 100,000 \geq -8,000 \quad (8)$$

Gurobi | Vaccine Allocation Problem



Problem Statement

A hospital has received a batch of 5000 COVID-19 vaccines. The hospital has 4 vaccine stations - Station A, Station B, Station C, and Station D. Each station has a priority based on the population it serves (e.g., elderly, children, chronically ill, disabled). The higher the priority value, the more vulnerable the population it serves. Additionally, each station has a storage capacity, and it is mandatory that every station receives at least 500 vaccines.

Gurobi | Vaccine Allocation Problem



Table 1: Station priorities and storage capacities.

Station	Priority	Storage Capacity (Vaccines)	Demand (Vaccines)
A	0.5	2300	2000
B	0.7	1700	2400
C	1.0	1200	1600
D	0.8	900	1000

Please formulate this problem as a linear program and find the optimal vaccine allocation strategy for this hospital using Gurobi.

Gurobi | Vaccine Allocation Problem



Mathematical Model

Decision variables:

X_i : number of vaccines allocated to station i , $\forall i \in \{A, B, C, D\}$

Parameters:

P_i : priority value of station i , $\forall i \in \{A, B, C, D\}$

S_i : storage capacity at station i , $\forall i \in \{A, B, C, D\}$

Gurobi | Vaccine Allocation Problem



Objective Function:

This function is designed to maximize the total “benefit” achieved from distributing the vaccines.

The benefit is calculated as the product of the number of vaccines allocated to each station and the priority of that station.

$$\text{Maximize} \quad \sum_{i \in \{A, B, C, D\}} P_i \cdot X_i$$

Gurobi | Vaccine Allocation Problem



Constraints:

1. Storage space constraints: This constraint ensures that the number of vaccines allocated to each station does not exceed the station's storage capacity. S_i is the storage capacity of station i .

$$X_i \leq S_i, \quad \forall i \in \{A, B, C, D\}$$

Gurobi | Vaccine Allocation Problem



2. Use all vaccines: This constraint guarantees that all available vaccines are allocated. The total number of vaccines available is 5000, so the sum of the vaccines allocated to all stations must equal this number.

$$\sum_{i \in \{A, B, C, D\}} X_i = 5000$$

Gurobi | Vaccine Allocation Problem



3. Minimum supply constraints: This constraint ensures that every station receives at least 500 vaccines. The decision variable X_i represents the number of vaccines allocated to each station, and i is the index of the stations.

Note: No need for non-negativity constraints.

$$X_i \geq 500, \quad \forall i \in \{A, B, C, D\}$$